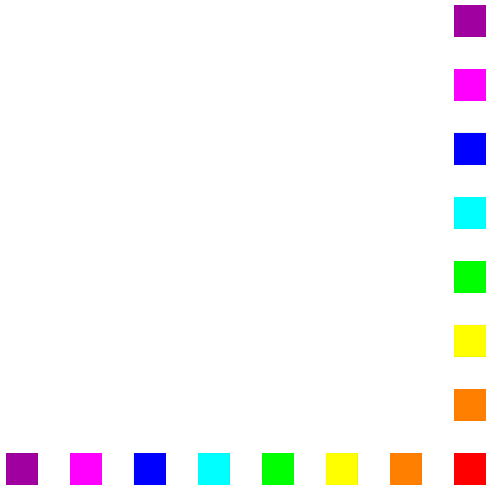
Introduction to Diagrammatic Stepwise Refinement

DIAGRAMMATIC STEPWISE REFINEMENT (DSR).

an approach to program design based on

The Program Structure Diagrams used in the Jackson Method (JSP)

Wirth's Stepwise Refinement approach to program design.



Steps in the DSR approach

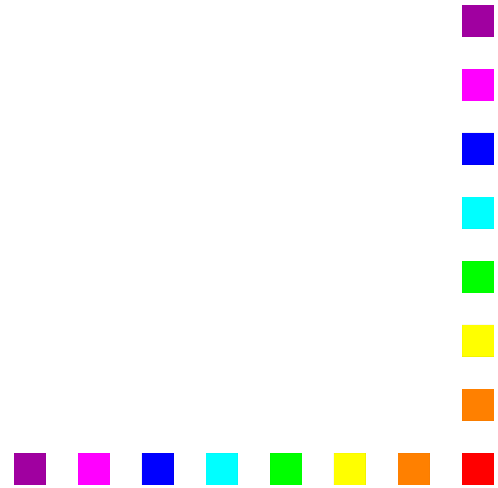
1. **Define the problem** completely.
2. **Work out what needs to be done to solve the problem.**
3. **Using Stepwise Refinement, design a Program Structure Diagram (PSD)** to represent your solution to the problem.
4. Write out the Executable Operations needed to produce the output from the input.
5. Assign the **Executable Operations** to the appropriate places in the PSD.
6. Insert the **Iteration** and **Selection** conditions.
7. Test your solution using test data.
8. Translate the populated PSD into code.

Stepwise Refinement

Is an approach to program design which involves breaking a task into sub-tasks.

Each sub-task is in turn broken into further sub-tasks.

This process continues until the sub-tasks are so simple that their implementation is straight forward.



Program Structure Diagrams

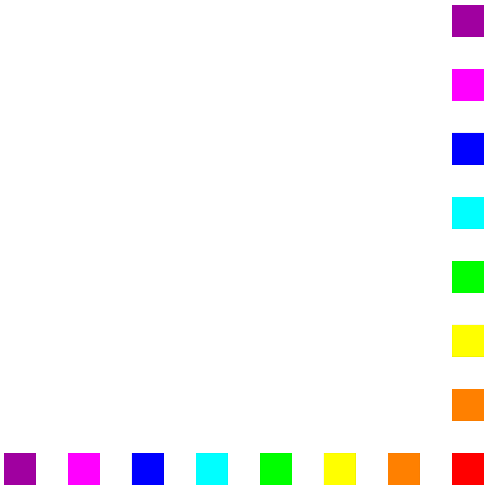
Program Structure Diagrams (PSDs) are used to **represent the hierarchy** of tasks and sub-tasks **created by Stepwise Refinement.**

They are drawn using the three classic constructs of Structured Programming.

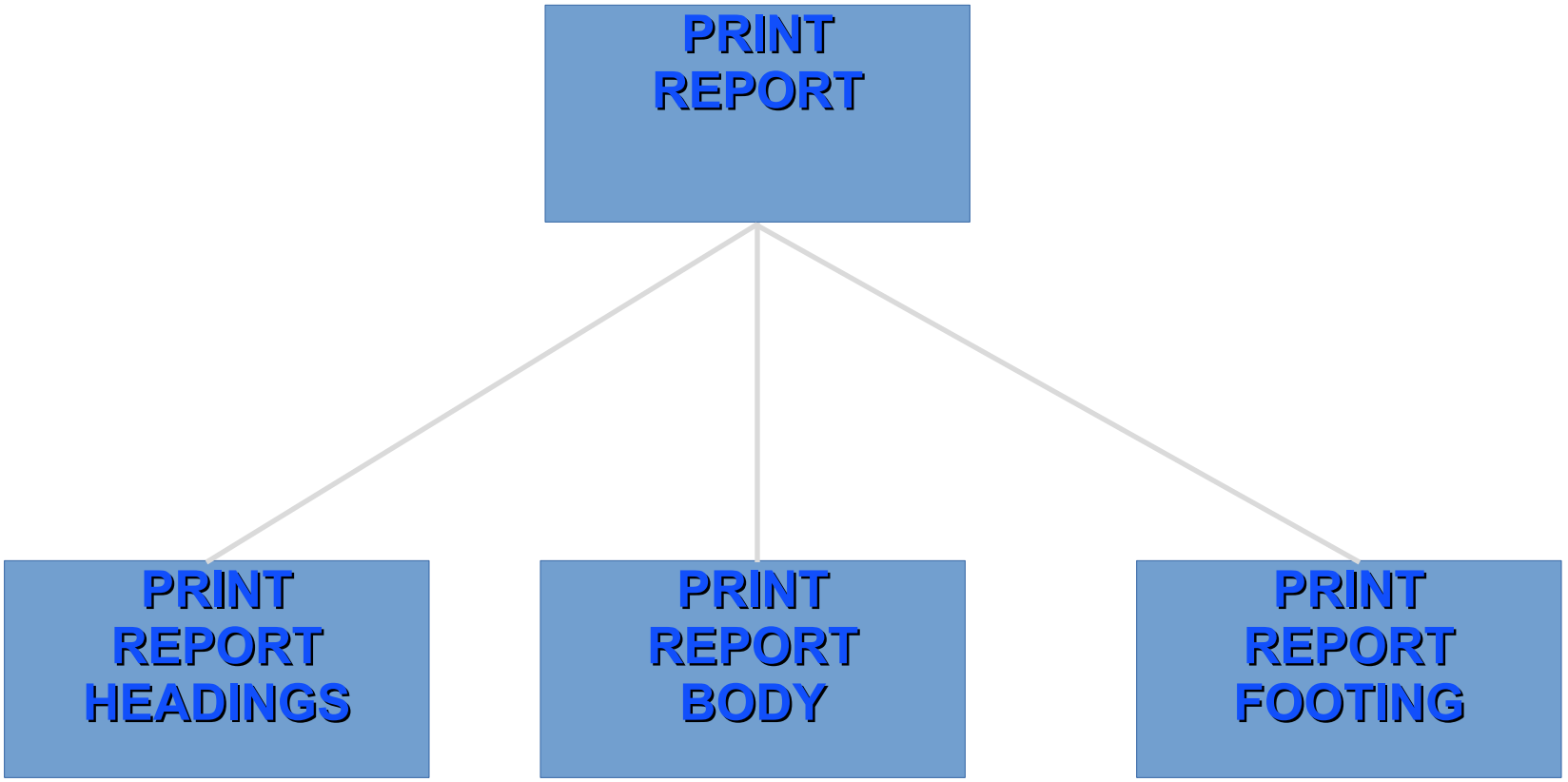
Sequence

Selection

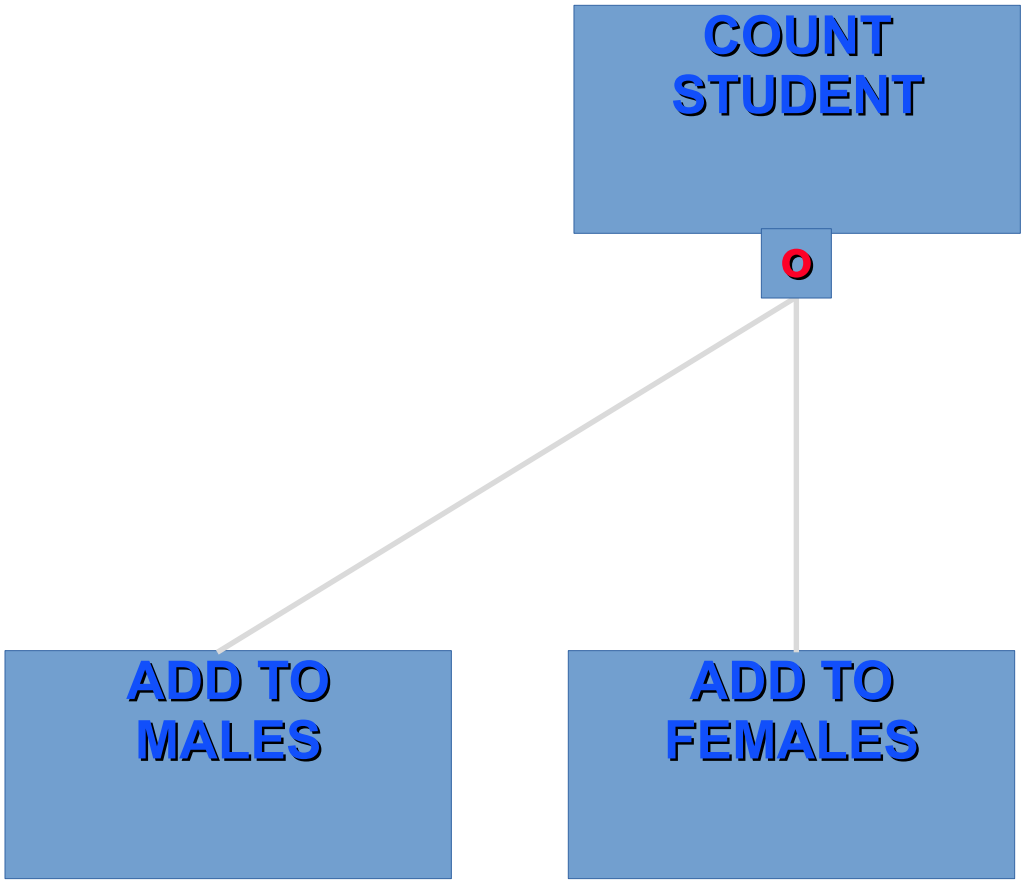
Iteration



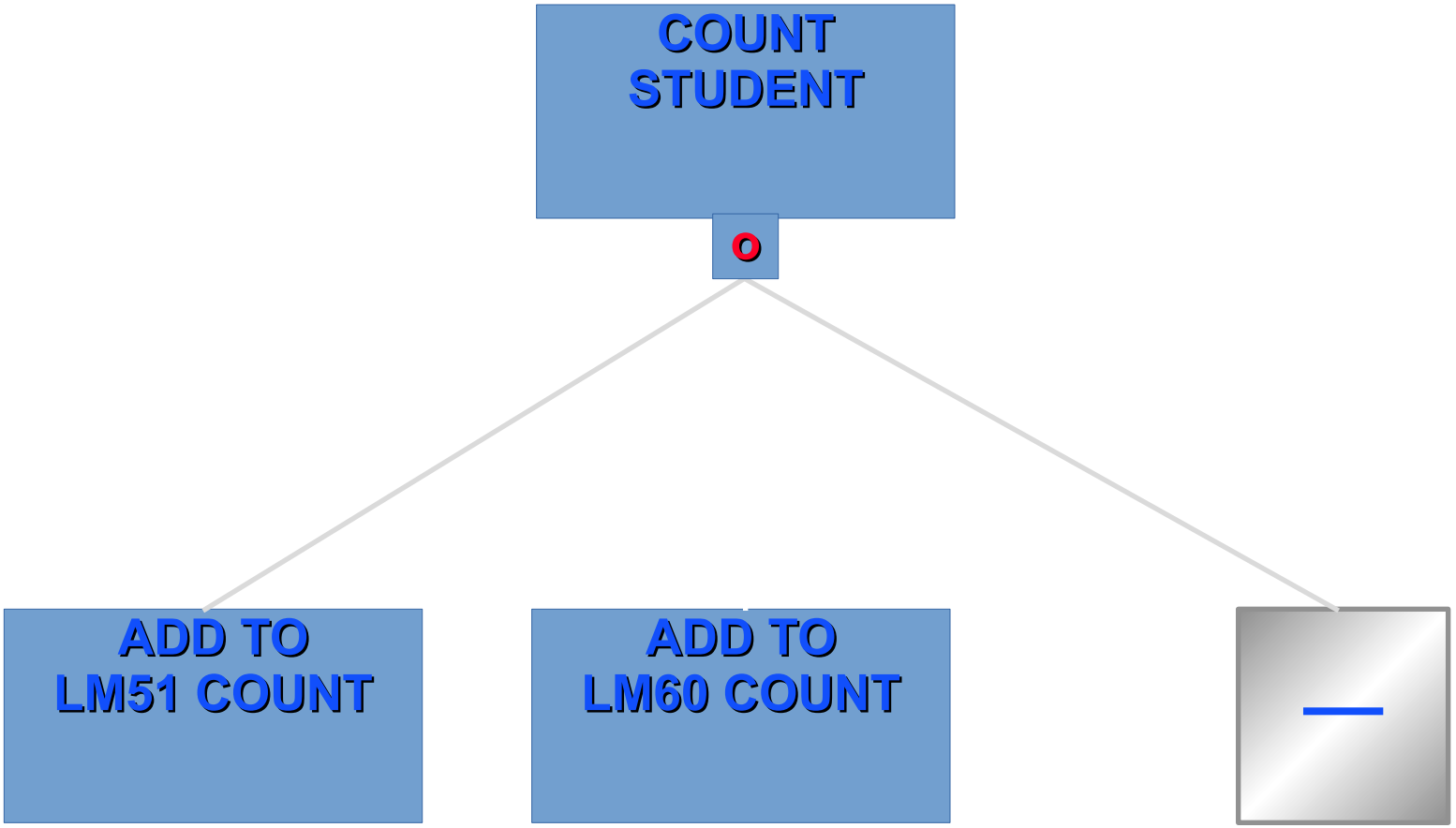
Sequence PSD



Selection PSD

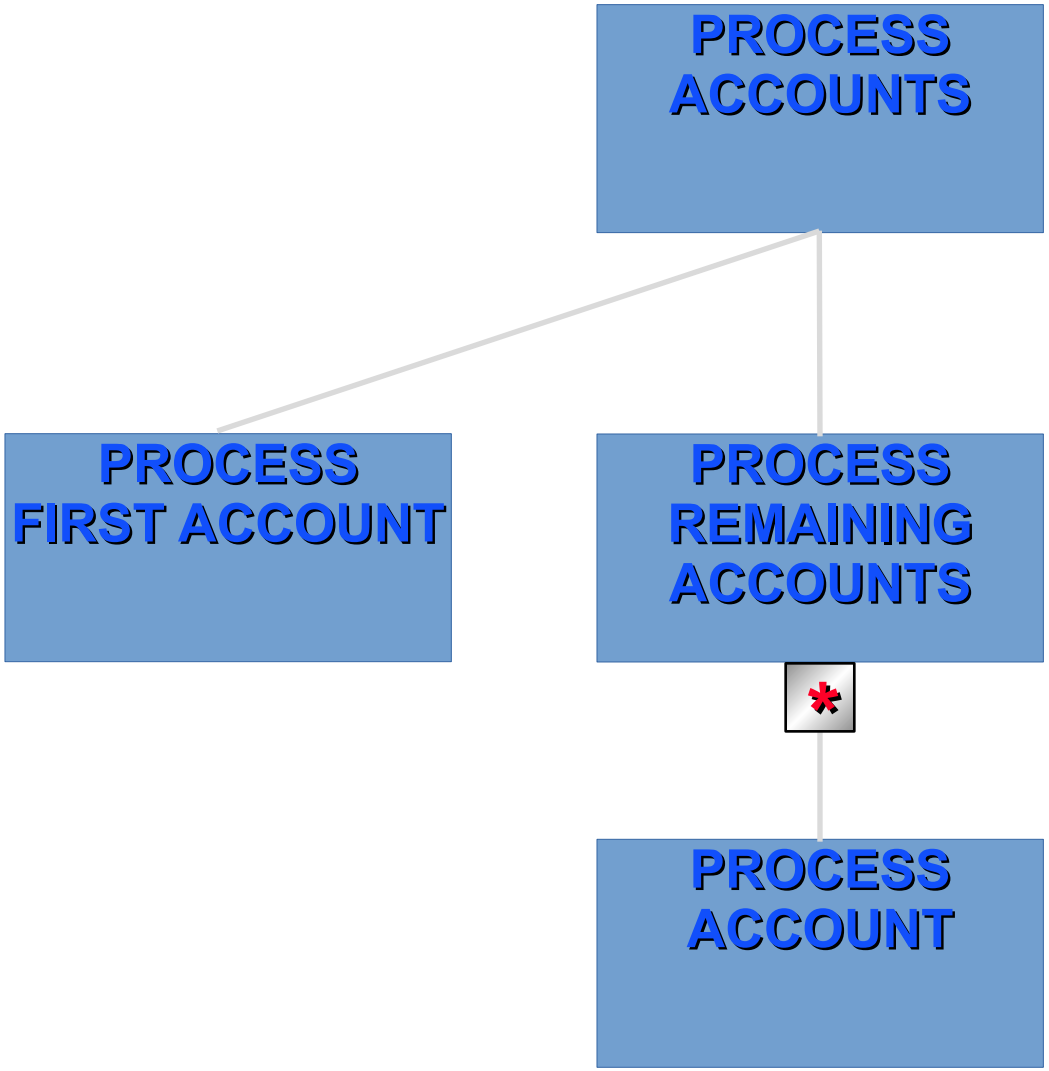


Selection with Null part

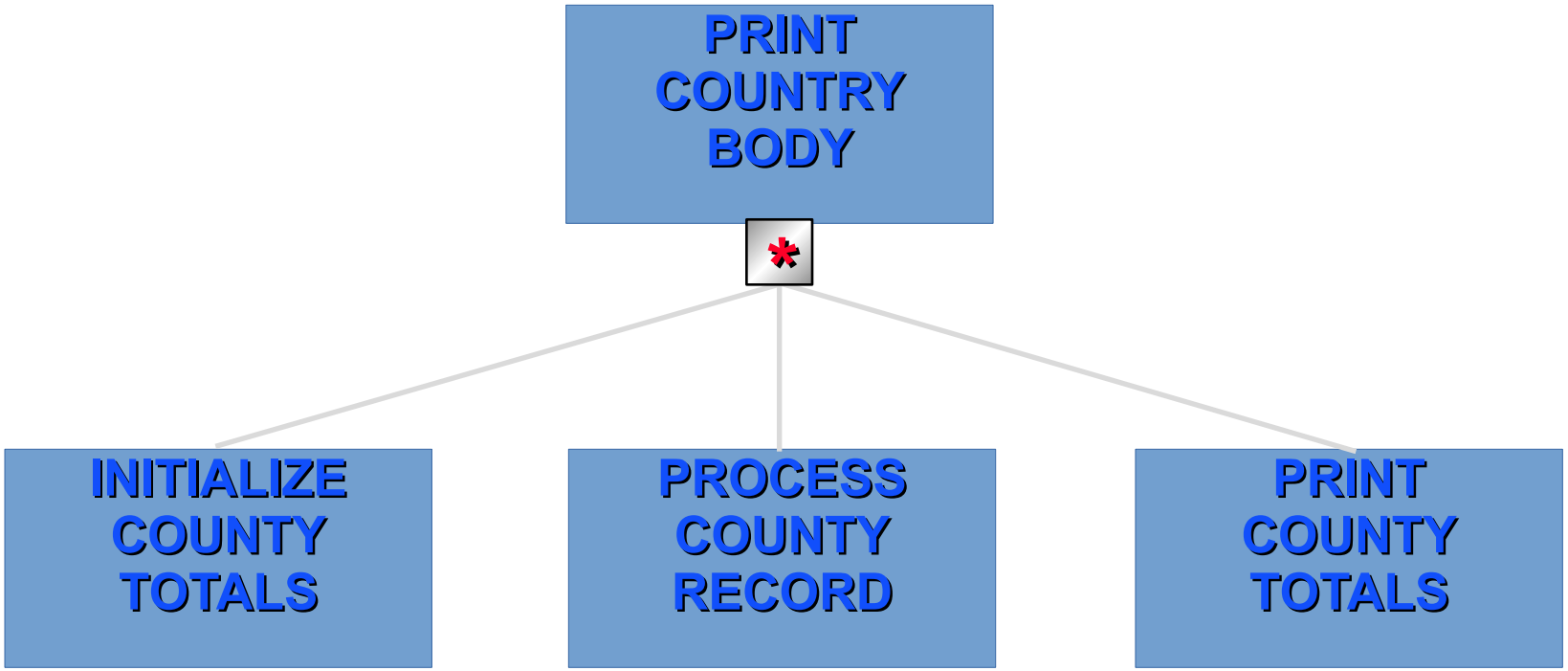


Iteration PSD

Iteration PSD - Explicit Repeat Loop



Iteration PSD

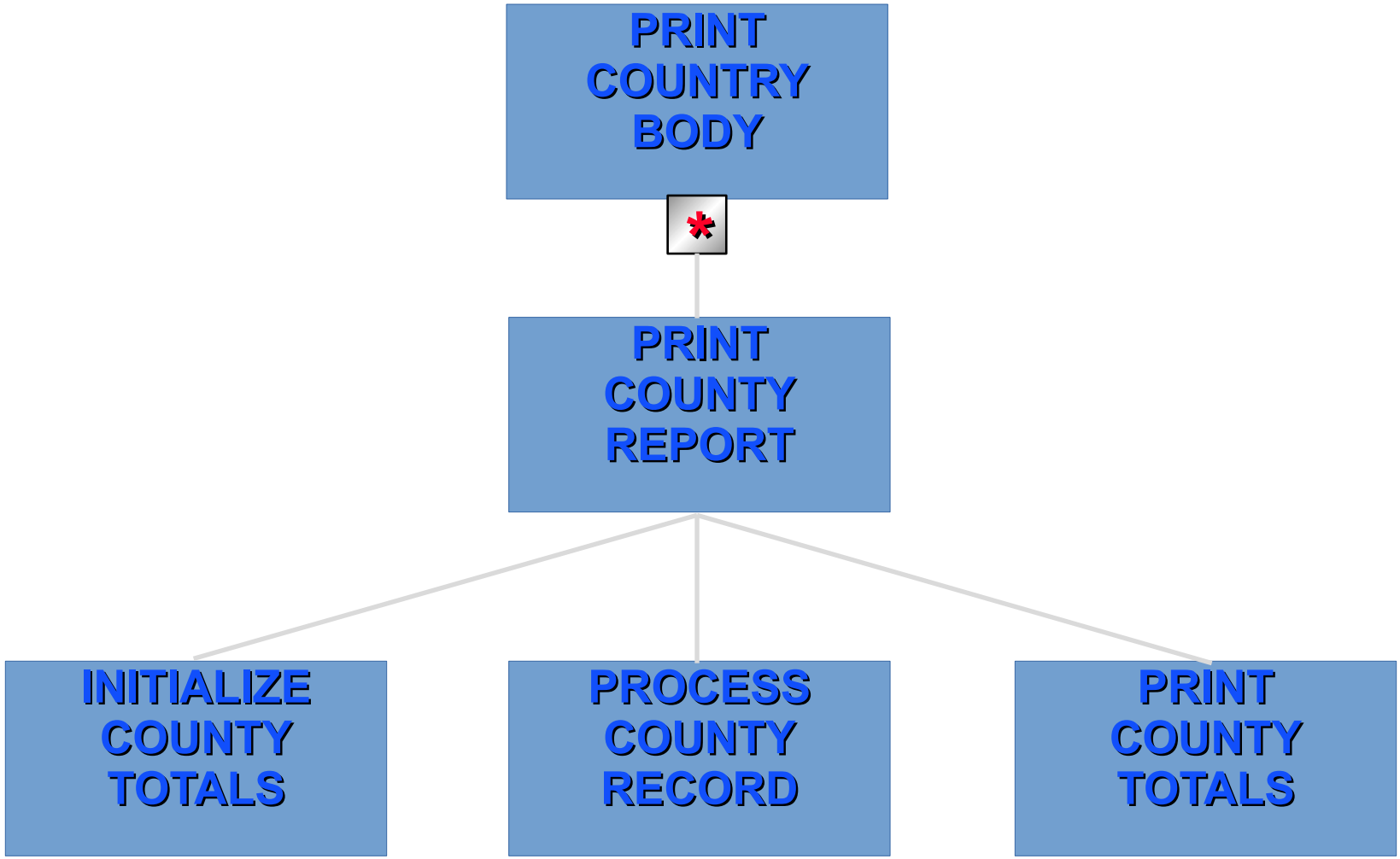


Print-Country-Body is an iteration of three components.

This is equivalent to ;

an iteration of one component which is in turn a sequence of three.

Iteration PSD



Payroll Proof Totals Program Specification

Write a program to produce proof totals for a weekly payroll. The payroll file consists of records with the following format:-

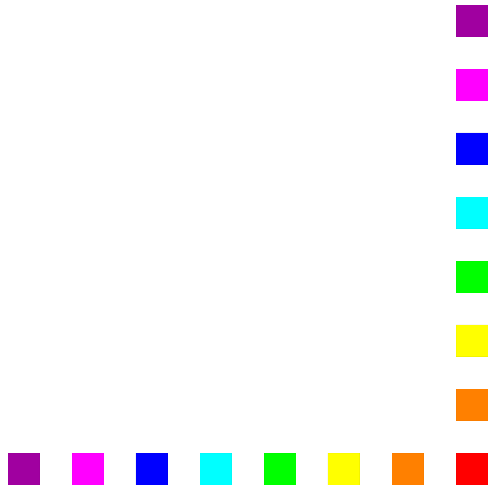
The print layout should be as follows:

PAYROLL PROOF TOTALS

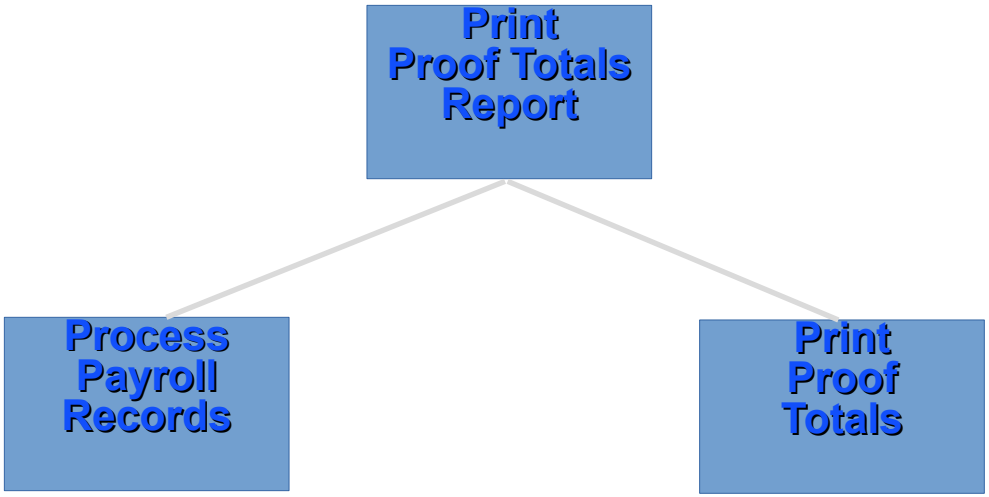
Regular Earnings :	**25**	**$2,111.23**
Overtime Earnings :	**2**	**$21.50**
Bonus Earnings :	**6**	**$123.45**

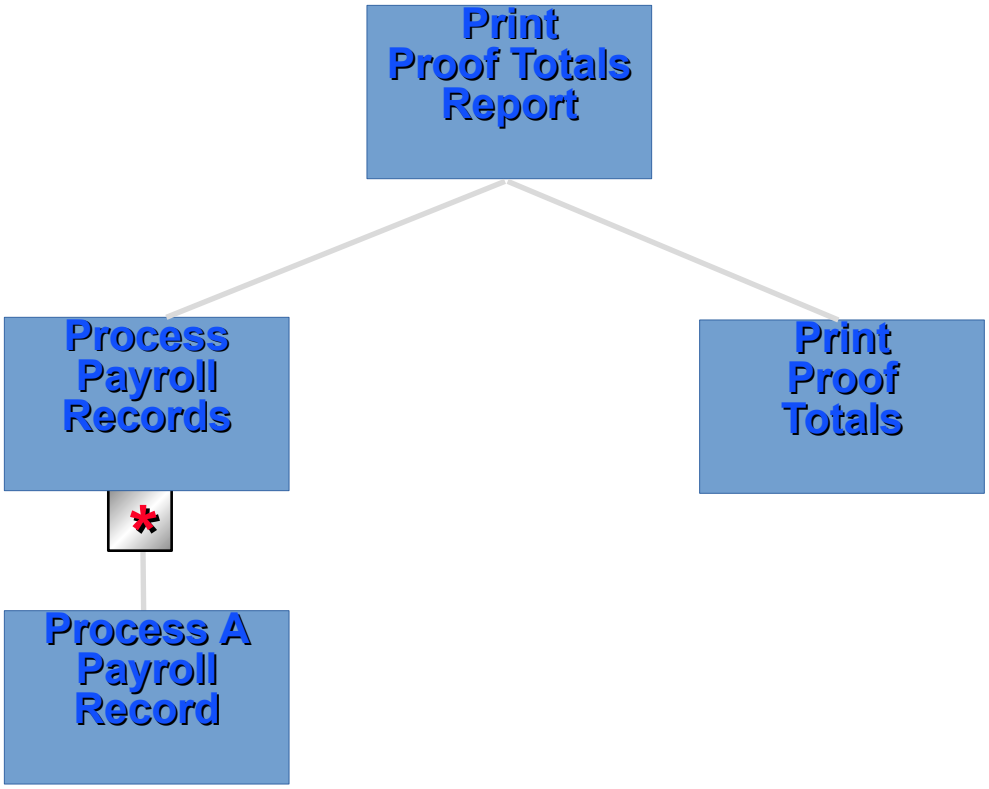
Applying DSR to the Payroll Proof Totals Program

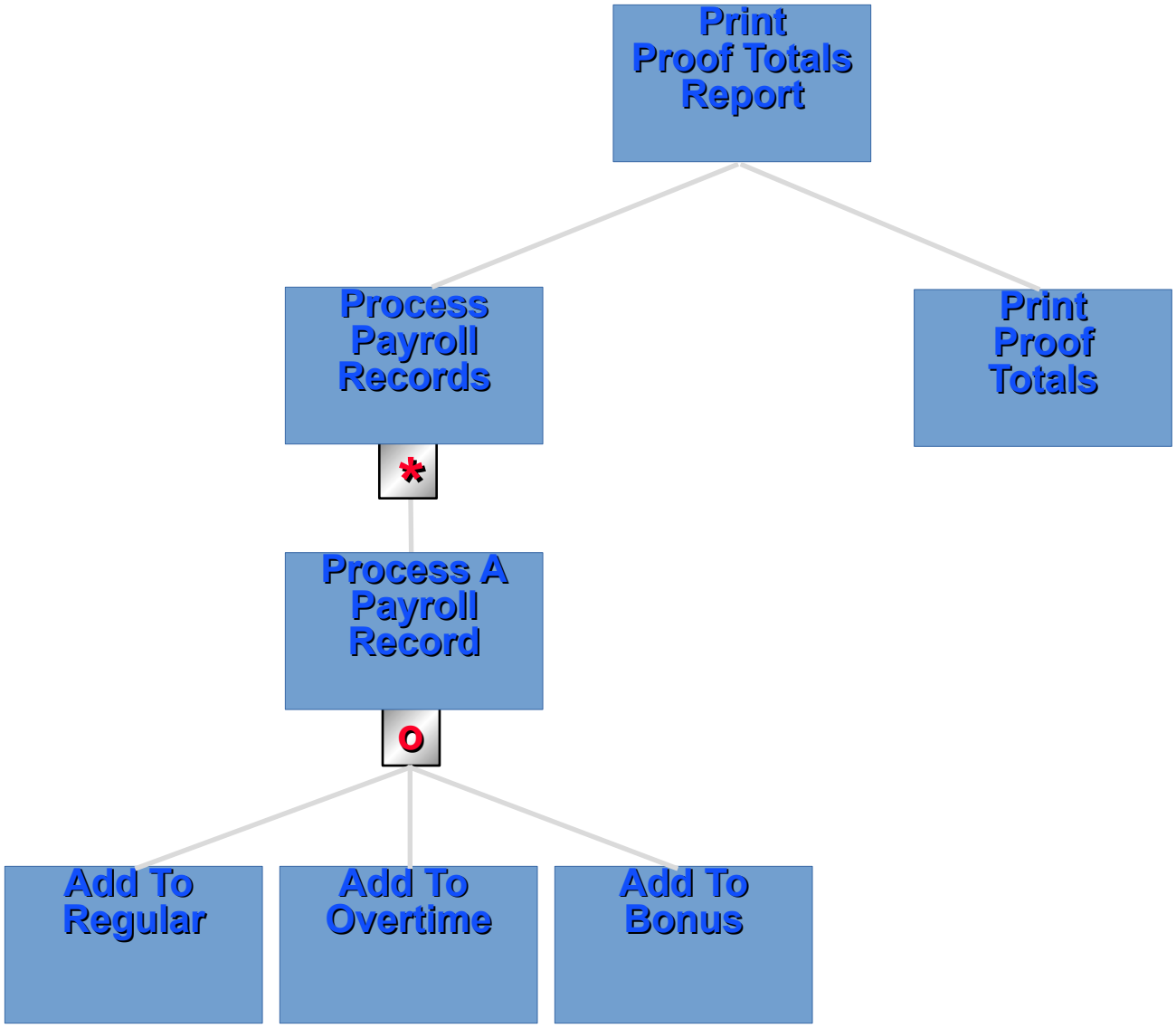
1. Define the problem completely
2. Work out what needs to be done to solve the problem.
3. Using Stepwise Refinement design a Program Structure Diagram (PSD) to represent your solution to the problem.

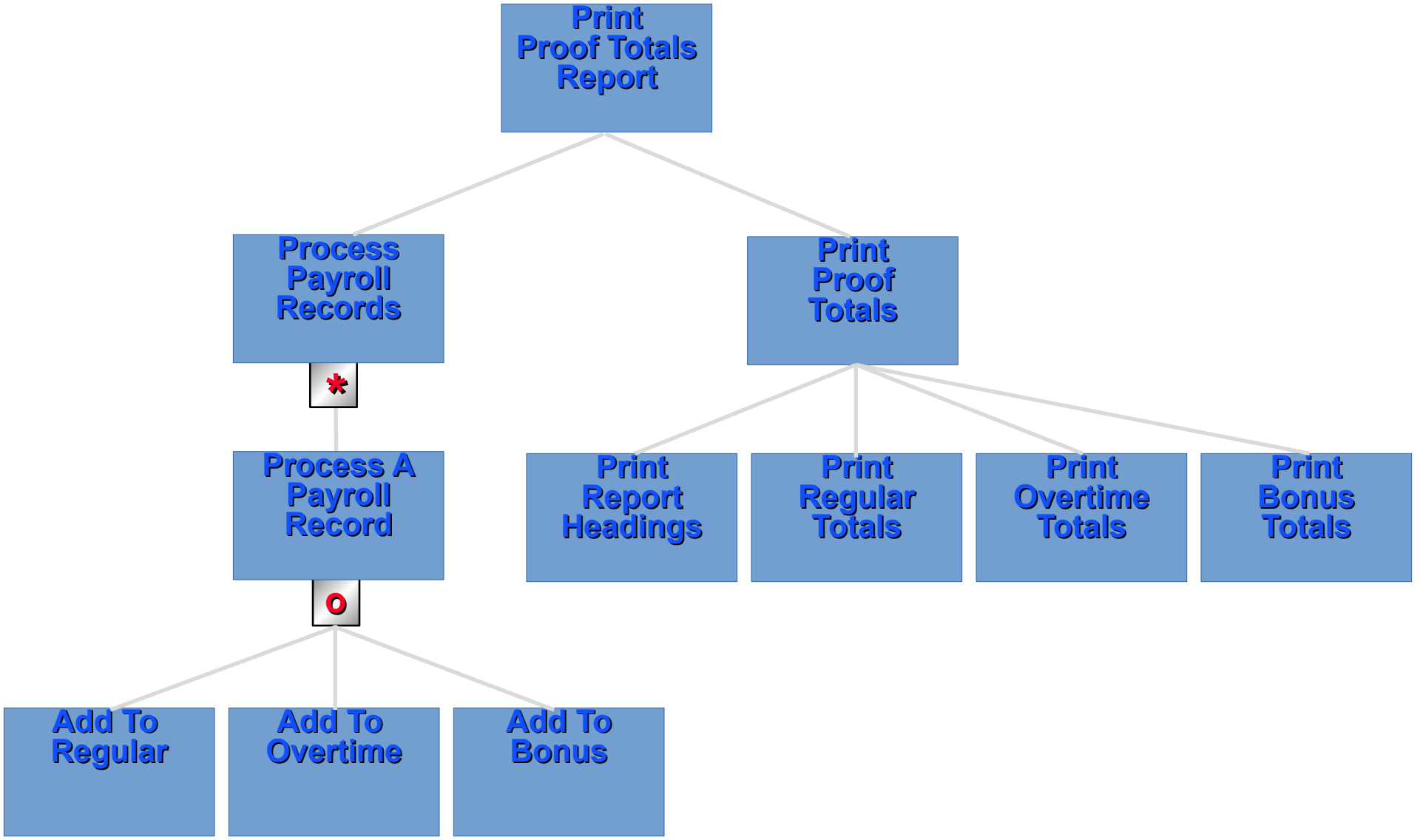



Print
Proof Totals
Report


Print
Proof Totals
Report
Process
Payroll
Records
Print
Proof
Totals


Print
Proof Totals
Report
Process
Payroll
Records
*
Process A
Payroll
Record
Print
Proof
Totals


Print Proof Totals Report
Process Payroll Records
Print Proof Totals
*
Process A Payroll Record
o
Add To Regular
Add To Overtime
Add To Bonus


Print Proof Totals Report
Process Payroll Records
*
Process A Payroll Record
o
Add To Regular
Add To Overtime
Add To Bonus
Print Proof Totals
Print Report Headings
Print Regular Totals
Print Overtime Totals
Print Bonus Totals

4. Write out the Executable Operations needed to produce the output from the input.

```
1.  OPEN OUTPUT Print-File.
2.  CLOSE Print-File.

3.  WRITE Print-Line FROM Heading-Line AFTER ADVANCING PAGE.
4.  WRITE Print-Line FROM Regular-Line AFTER ADVANCING 1 LINE.
5.  WRITE Print-Line FROM Overtime-Line AFTER ADVANCING 1 LINE.
6.  WRITE Print-Line FROM Bonus-Line AFTER ADVANCING 1 LINE.

7.  MOVE Regular-Total TO Print-Reg-Total.
8.  MOVE Overtime-Total TO Print-Overtime-Total.
9.  MOVE Bonus-Total TO Print-Bonus-Total.

10. MOVE Regular-Count TO Print-Reg-Count.
11. MOVE Overtime-Count TO Print-Over-Count.
12. MOVE Bonus-Count TO Print-Bonus-Count.

13. ADD Earnings TO Regular-Total.
14. ADD Earnings TO Overtime-Total.
15. ADD Earnings TO Bonus-Total.

16. ADD 1 TO Regular-Count.
17. ADD 1 TO Overtime-Count.
18. ADD 1 TO Bonus-Count.

19. READ Payroll-File
        AT END SET End-of-File TO TRUE
    END-READ.

20. OPEN INPUT Payroll-File.
21. CLOSE Payroll-File.
```

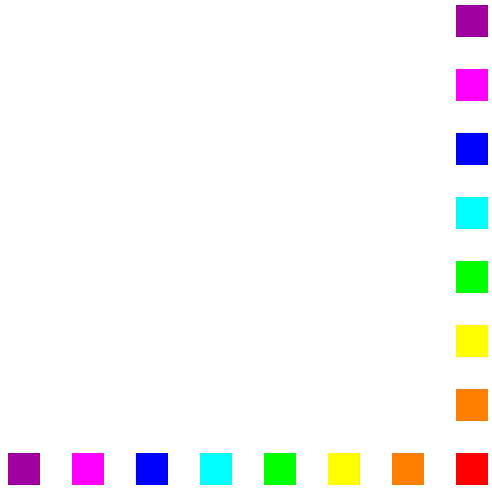
5. Assign the Executable Operations to the appropriate places in the PSD.

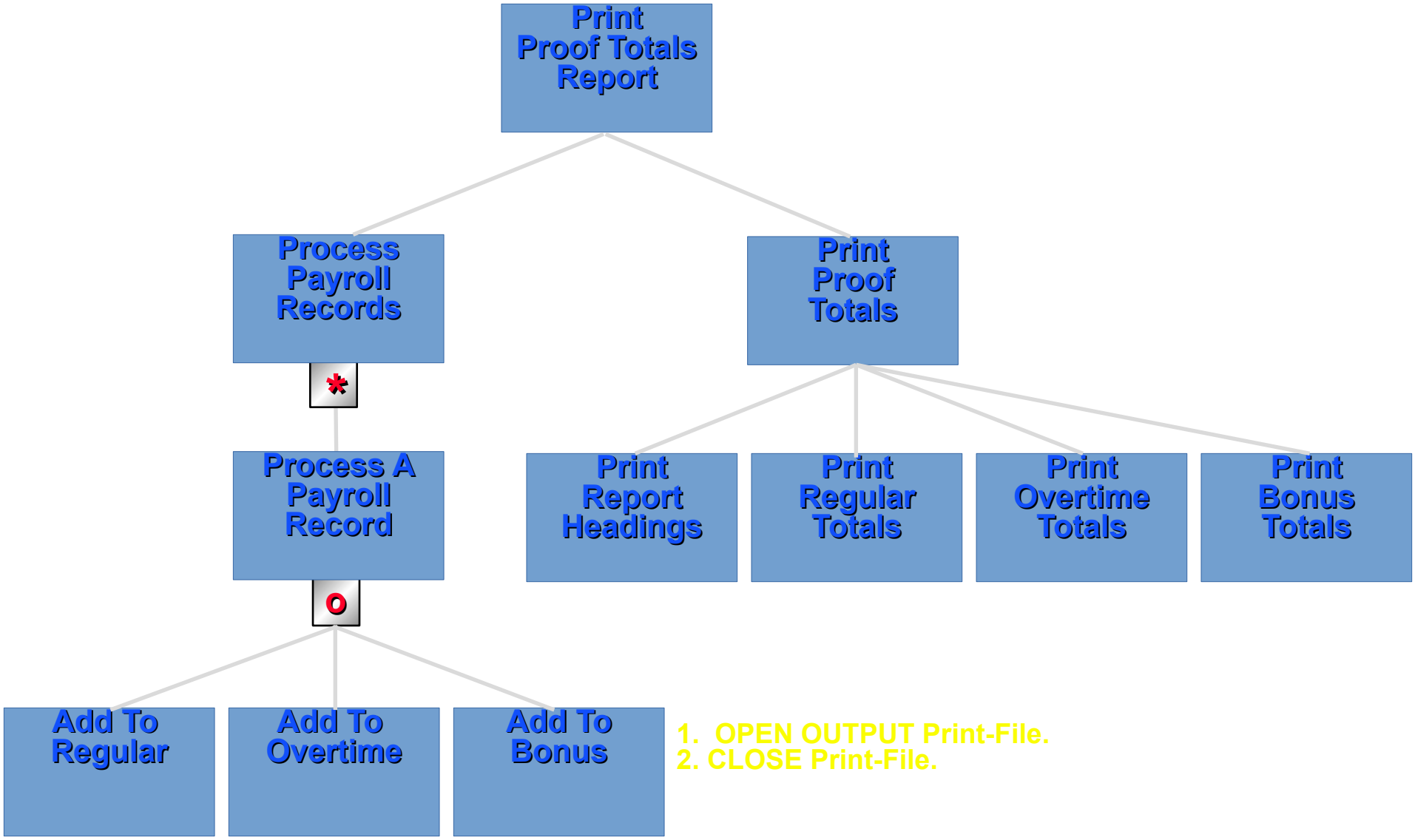
Algorithm

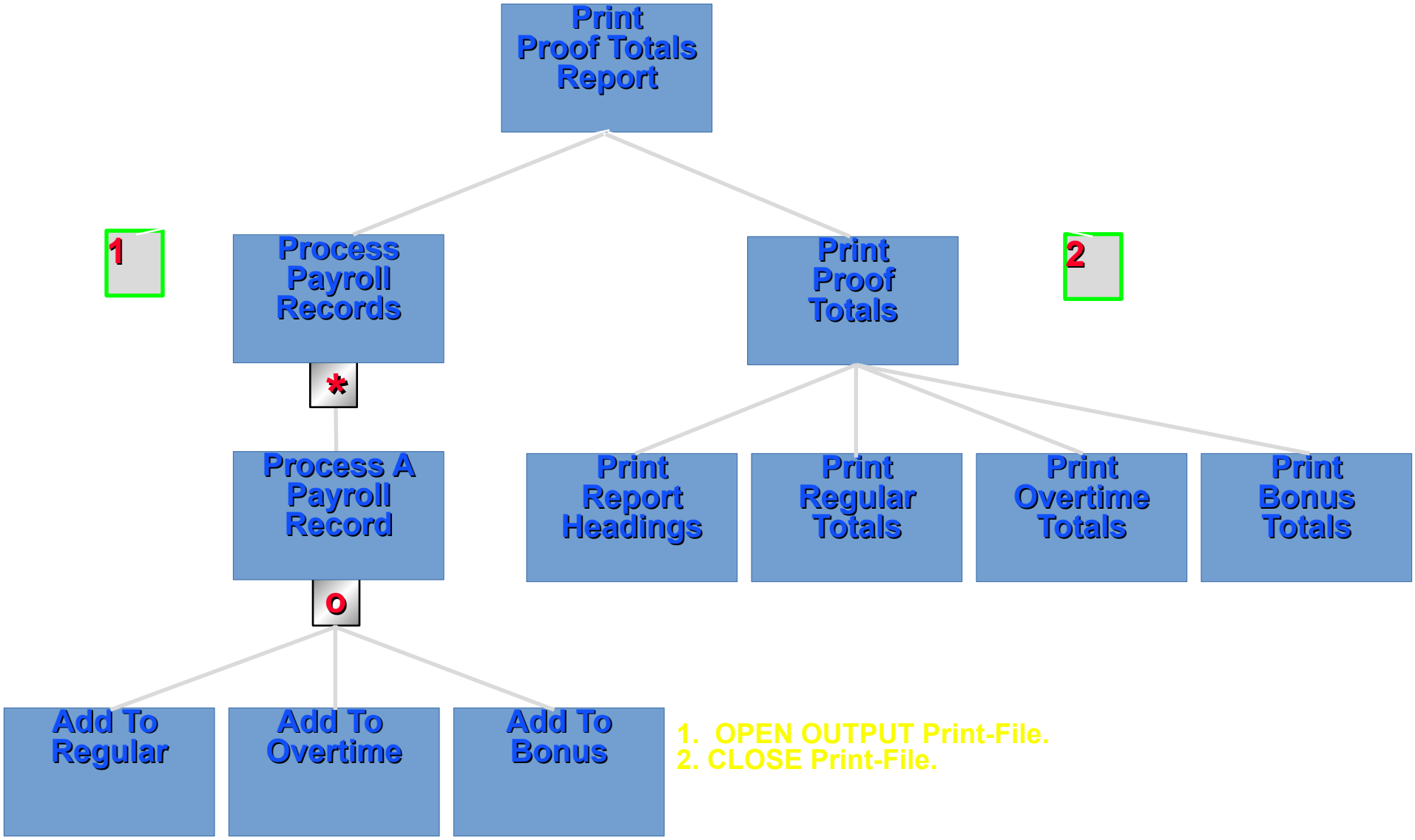
For all Executable Operations.

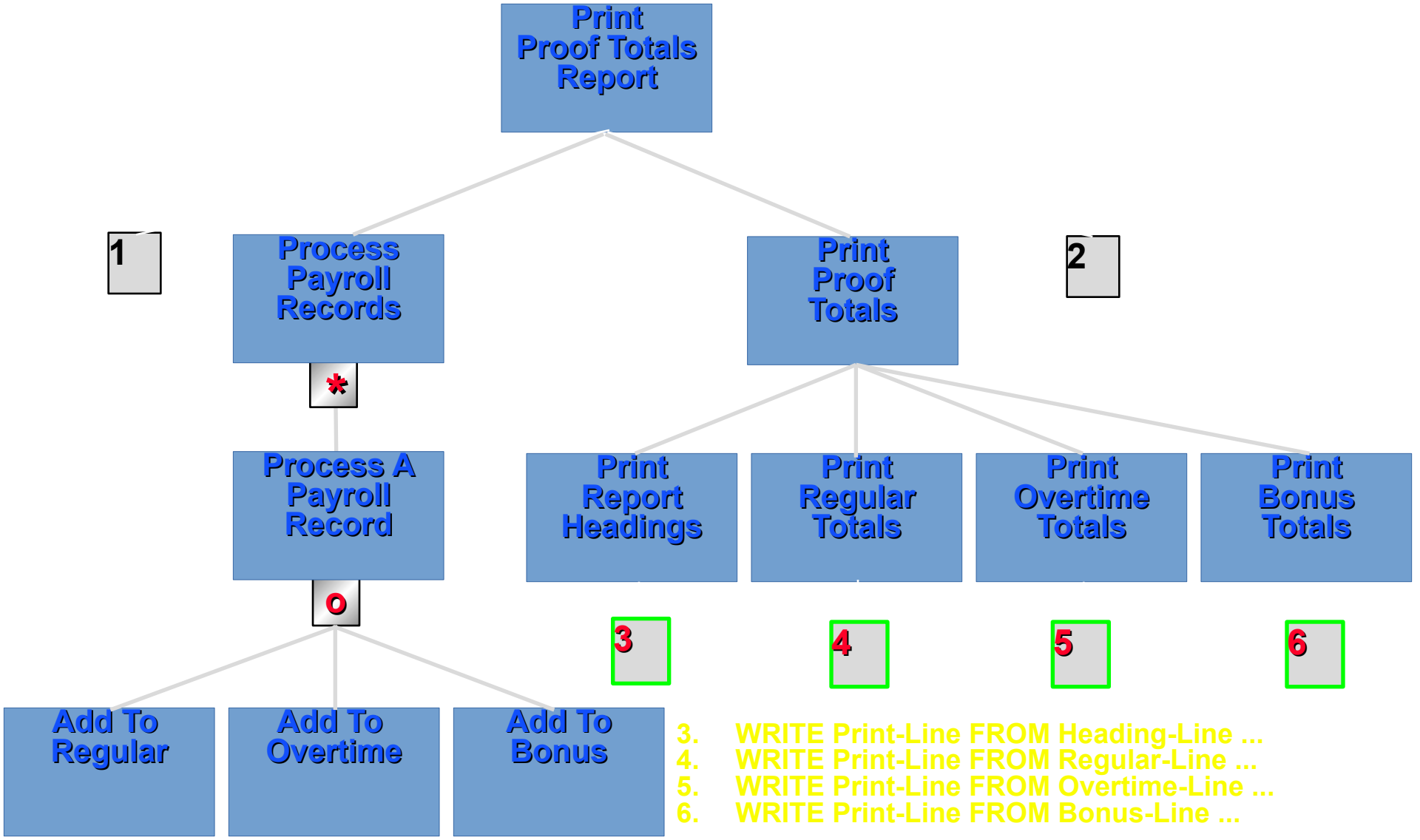
Get an Executable Oper

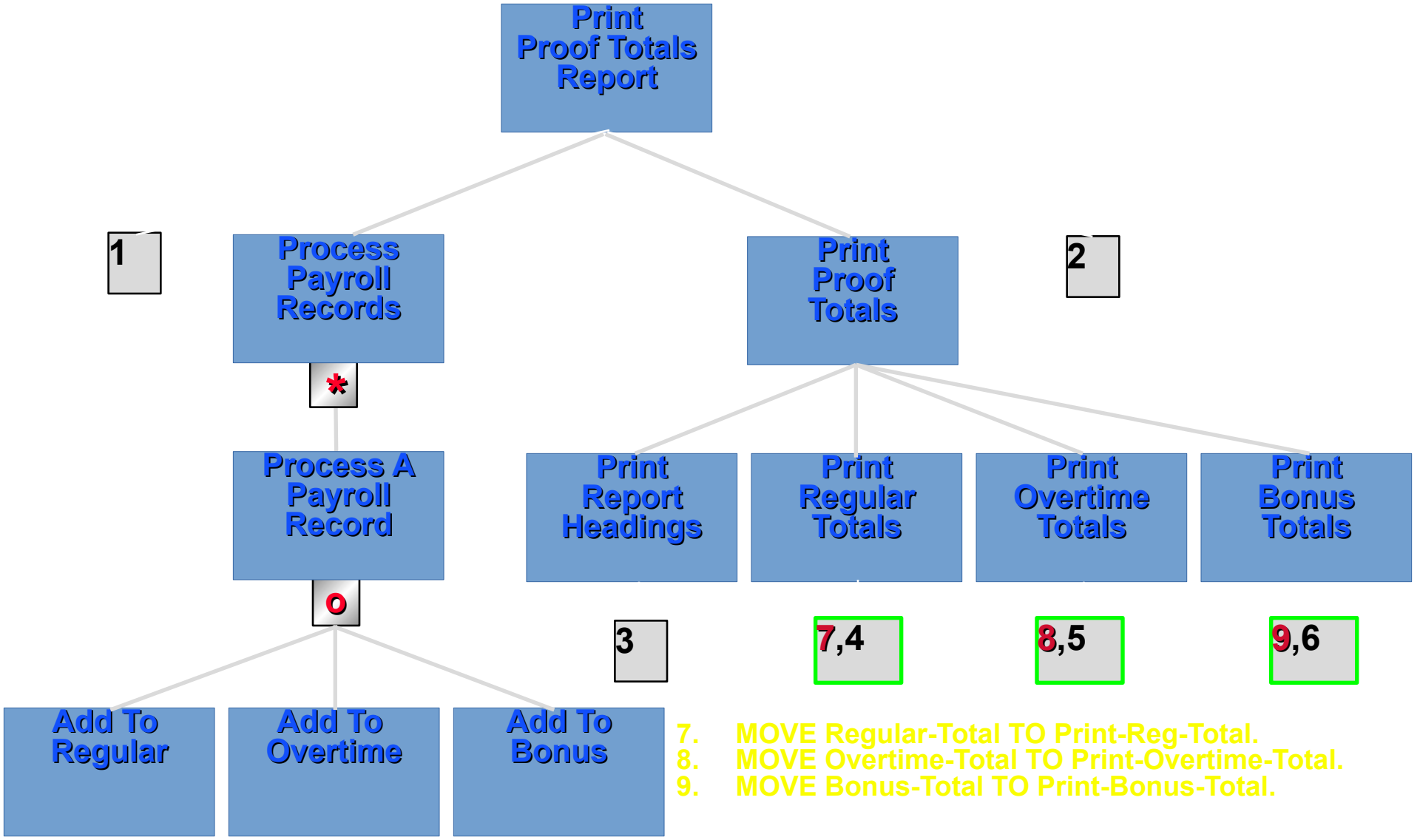
Ask "To what componer
attach this operation
work?".

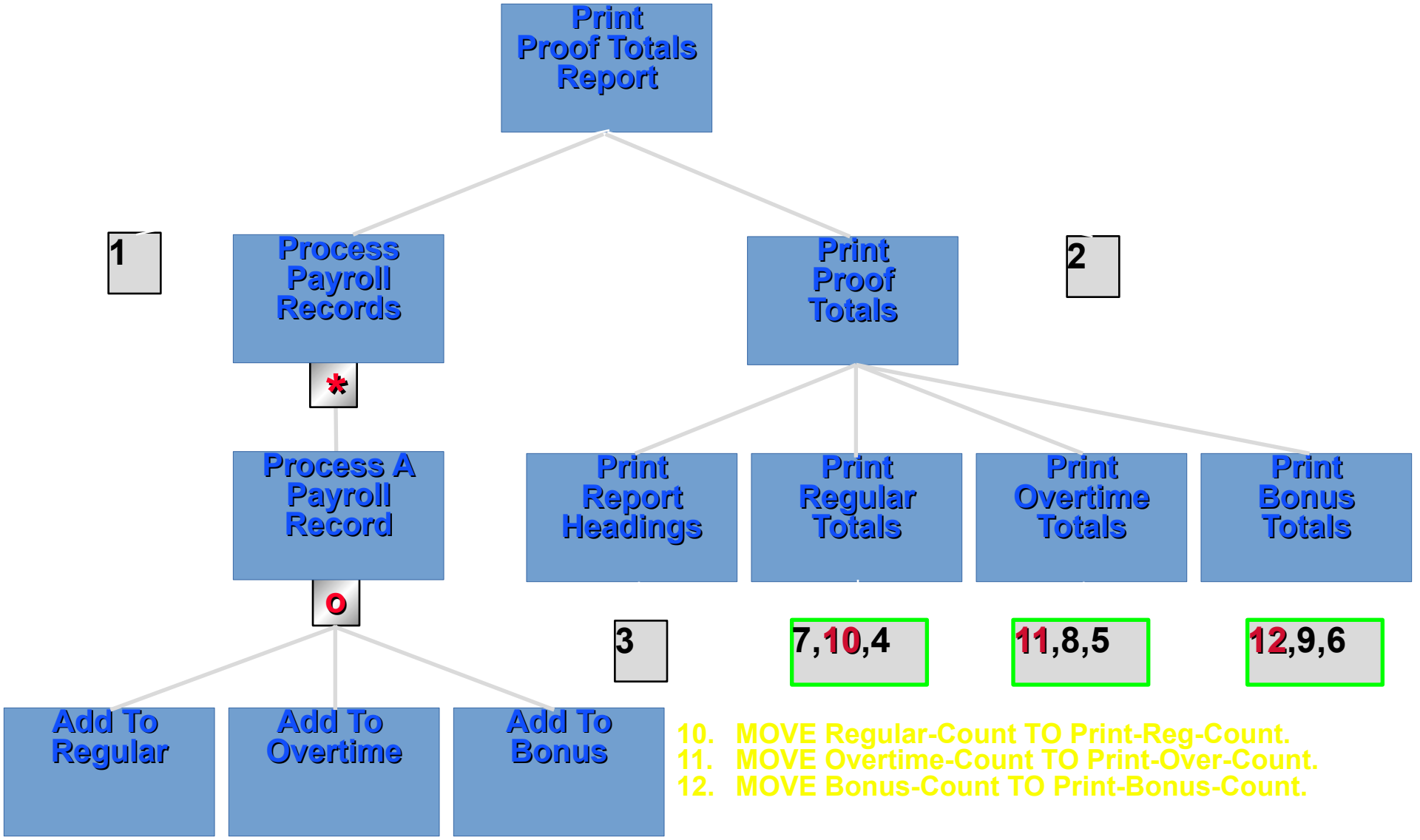


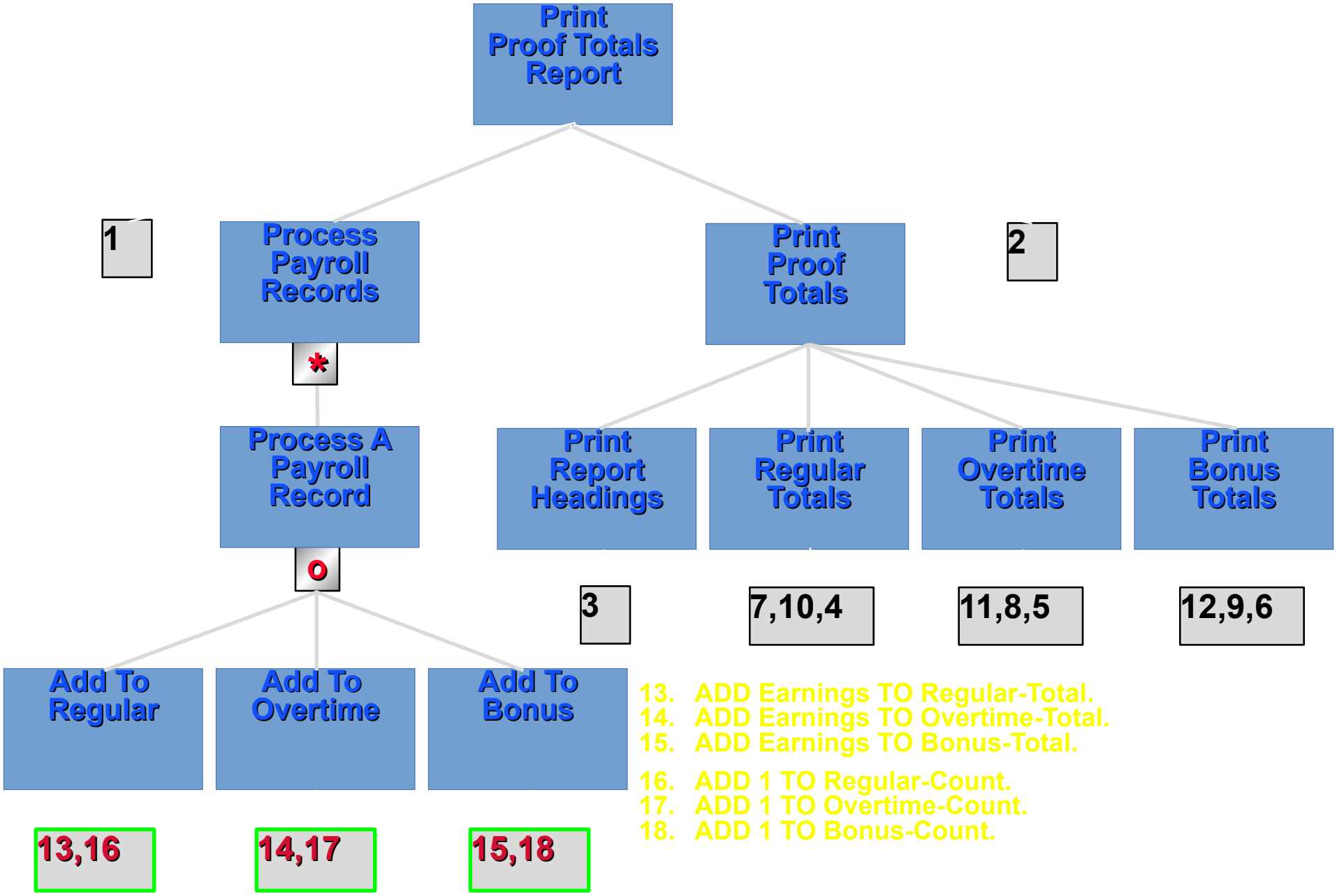

Print Proof Totals Report
Process Payroll Records
*
Process A Payroll Record
o
Add To Regular
Add To Overtime
Add To Bonus
Print Proof Totals
Print Report Headings
Print Regular Totals
Print Overtime Totals
Print Bonus Totals
1. OPEN OUTPUT Print-File.
2. CLOSE Print-File.

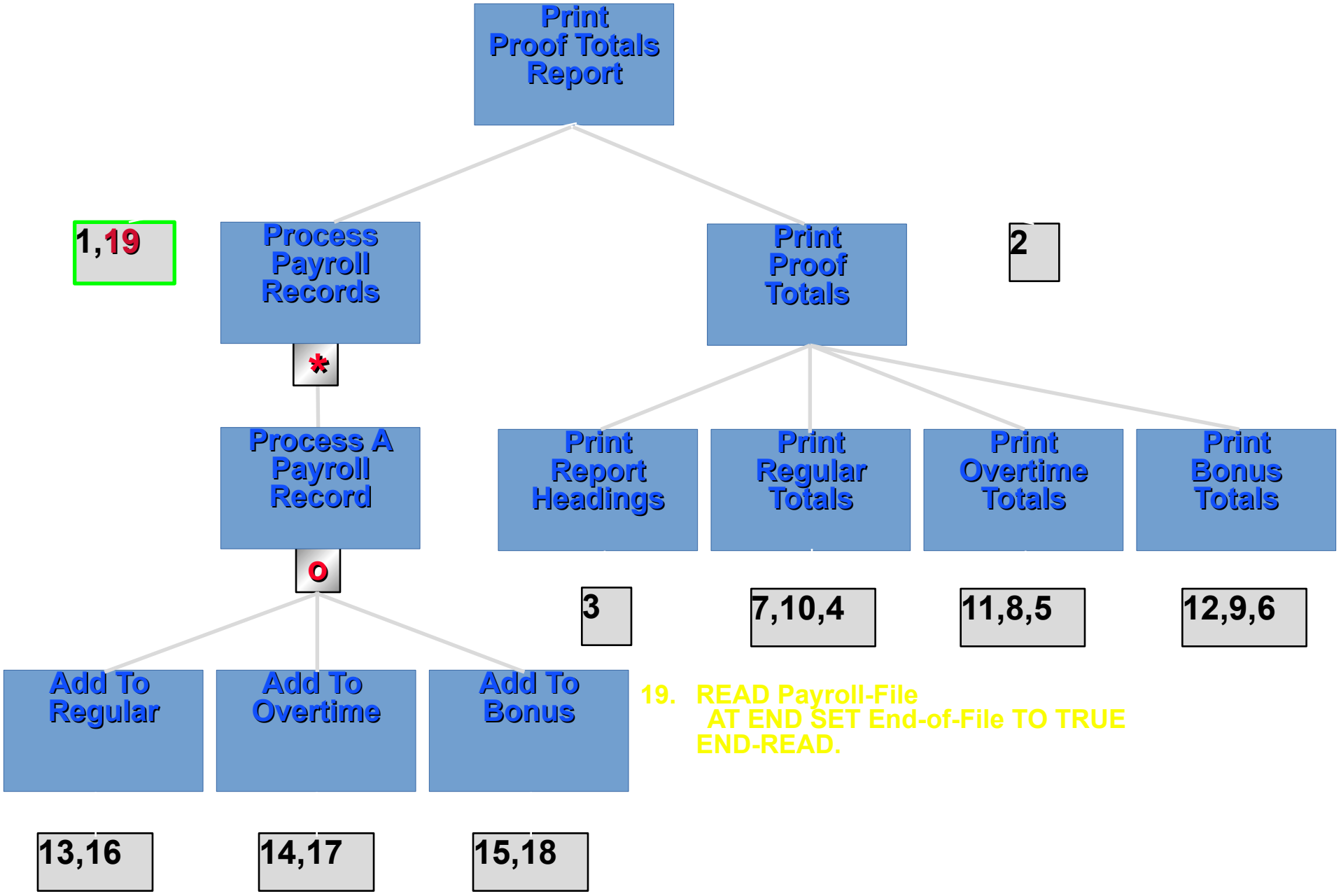

Print
Proof Totals
Report
1
Process
Payroll
Records
*
Process A
Payroll
Record
o
Add To
Regular
Add To
Overtime
Add To
Bonus
Print
Proof
Totals
2
Print
Report
Headings
Print
Regular
Totals
Print
Overtime
Totals
Print
Bonus
Totals
1. OPEN OUTPUT Print-File.
2. CLOSE Print-File.

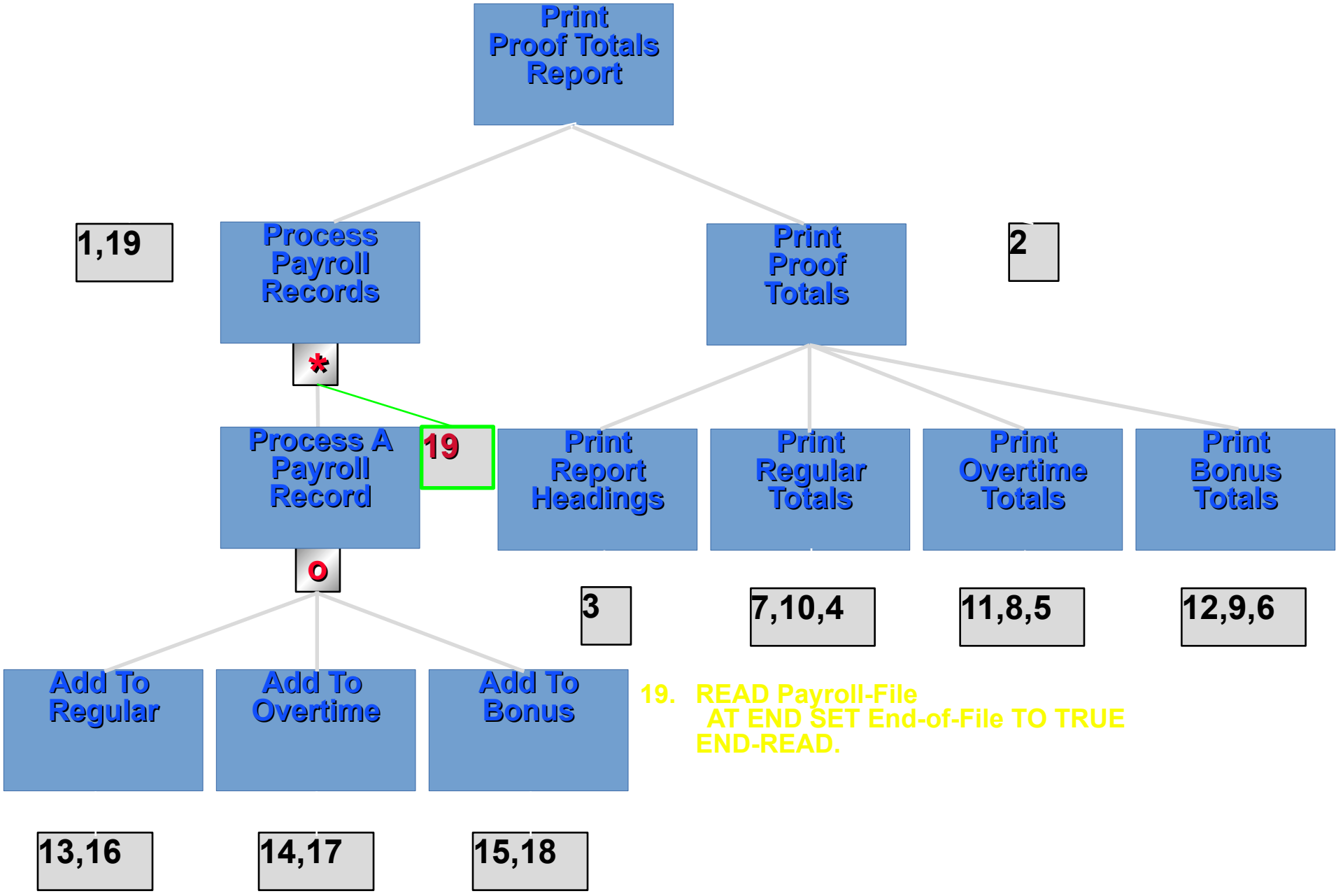

Print Proof Totals Report
1
Process Payroll Records
*
Process A Payroll Record
o
Add To Regular
Add To Overtime
Add To Bonus
2
Print Proof Totals
Print Report Headings
Print Regular Totals
Print Overtime Totals
Print Bonus Totals
3
4
5
6
3. WRITE Print-Line FROM Heading-Line ...
4. WRITE Print-Line FROM Regular-Line ...
5. WRITE Print-Line FROM Overtime-Line ...
6. WRITE Print-Line FROM Bonus-Line ...

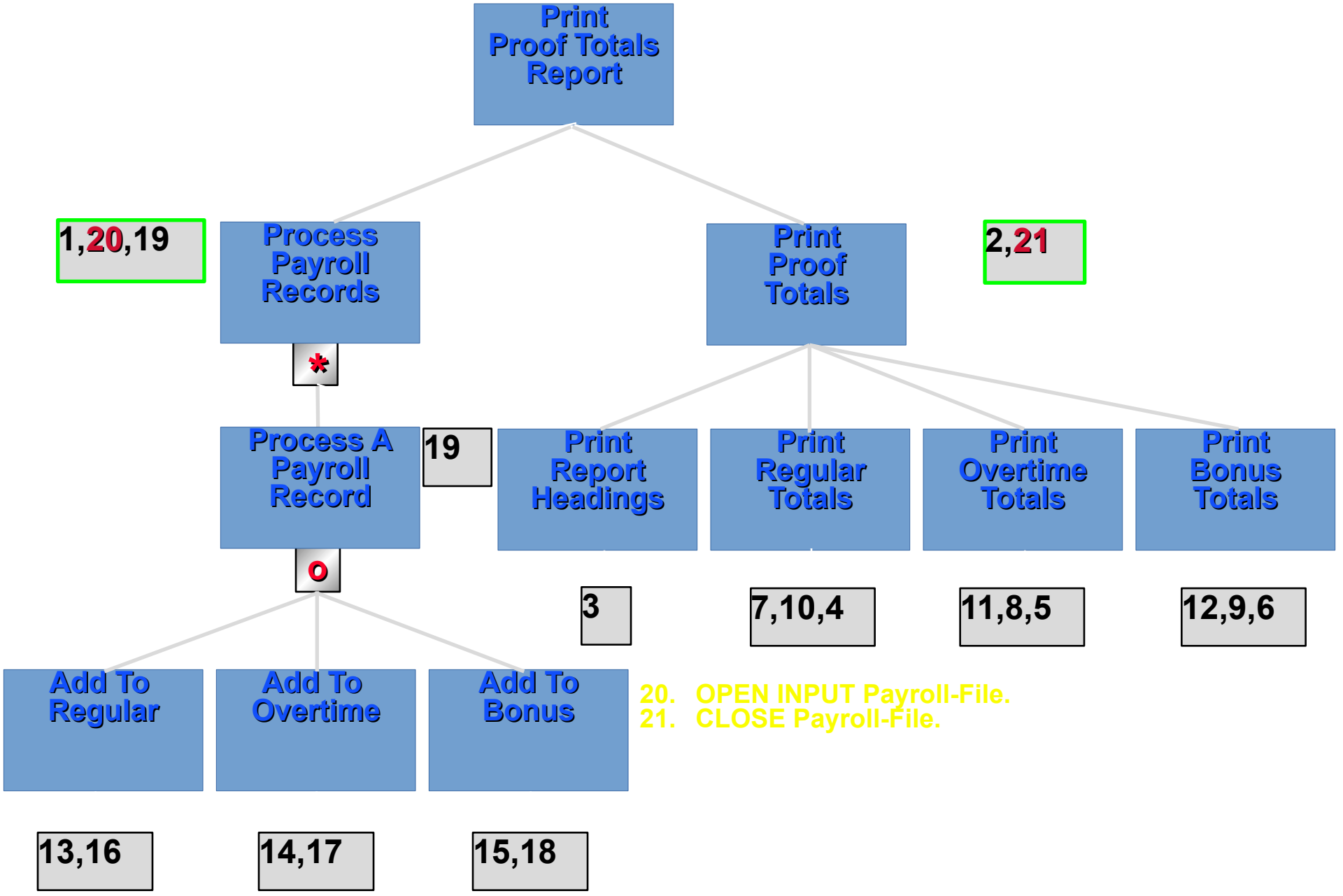

Print Proof Totals Report
1
Process Payroll Records
*
Process A Payroll Record
o
Add To Regular
Add To Overtime
Add To Bonus
2
Print Proof Totals
Print Report Headings
Print Regular Totals
Print Overtime Totals
Print Bonus Totals
3
7,4
8,5
9,6
7. MOVE Regular-Total TO Print-Reg-Total.
8. MOVE Overtime-Total TO Print-Overtime-Total.
9. MOVE Bonus-Total TO Print-Bonus-Total.


Print Proof Totals Report
1
Process Payroll Records
*
Process A Payroll Record
o
Add To Regular
Add To Overtime
Add To Bonus
2
Print Proof Totals
Print Report Headings
Print Regular Totals
Print Overtime Totals
Print Bonus Totals
3
7,10,4
11,8,5
12,9,6
10. MOVE Regular-Count TO Print-Reg-Count.
11. MOVE Overtime-Count TO Print-Over-Count.
12. MOVE Bonus-Count TO Print-Bonus-Count.


Print Proof Totals Report
1
Process Payroll Records
*
Process A Payroll Record
o
Add To Regular
Add To Overtime
Add To Bonus
13,16
14,17
15,18
2
Print Proof Totals
Print Report Headings
Print Regular Totals
Print Overtime Totals
Print Bonus Totals
3
7,10,4
11,8,5
12,9,6
13. ADD Earnings TO Regular-Total.
14. ADD Earnings TO Overtime-Total.
15. ADD Earnings TO Bonus-Total.
16. ADD 1 TO Regular-Count.
17. ADD 1 TO Overtime-Count.
18. ADD 1 TO Bonus-Count.


Print Proof Totals Report
1,19
Process Payroll Records
*
Process A Payroll Record
o
Add To Regular
Add To Overtime
Add To Bonus
13,16
14,17
15,18
Print Proof Totals
2
Print Report Headings
3
Print Regular Totals
7,10,4
Print Overtime Totals
11,8,5
Print Bonus Totals
12,9,6
19. READ Payroll-File
AT END SET End-of-File TO TRUE
END-READ.


Print Proof Totals Report
1,19
Process Payroll Records
*
Process A Payroll Record
19
o
Add To Regular
Add To Overtime
Add To Bonus
13,16
14,17
15,18
Print Proof Totals
2
Print Report Headings
3
Print Regular Totals
7,10,4
Print Overtime Totals
11,8,5
Print Bonus Totals
12,9,6
19. READ Payroll-File
AT END SET End-of-File TO TRUE
END-READ.


Print Proof Totals Report
1,20,19
Process Payroll Records
*
Process A Payroll Record
19
o
Add To Regular
Add To Overtime
Add To Bonus
13,16
14,17
15,18
Print Proof Totals
2,21
Print Report Headings
Print Regular Totals
Print Overtime Totals
Print Bonus Totals
3
7,10,4
11,8,5
12,9,6
20. OPEN INPUT Payroll-File.
21. CLOSE Payroll-File.

6. Insert the Iteration and Selection conditions.

Go to each iteration component and ask;

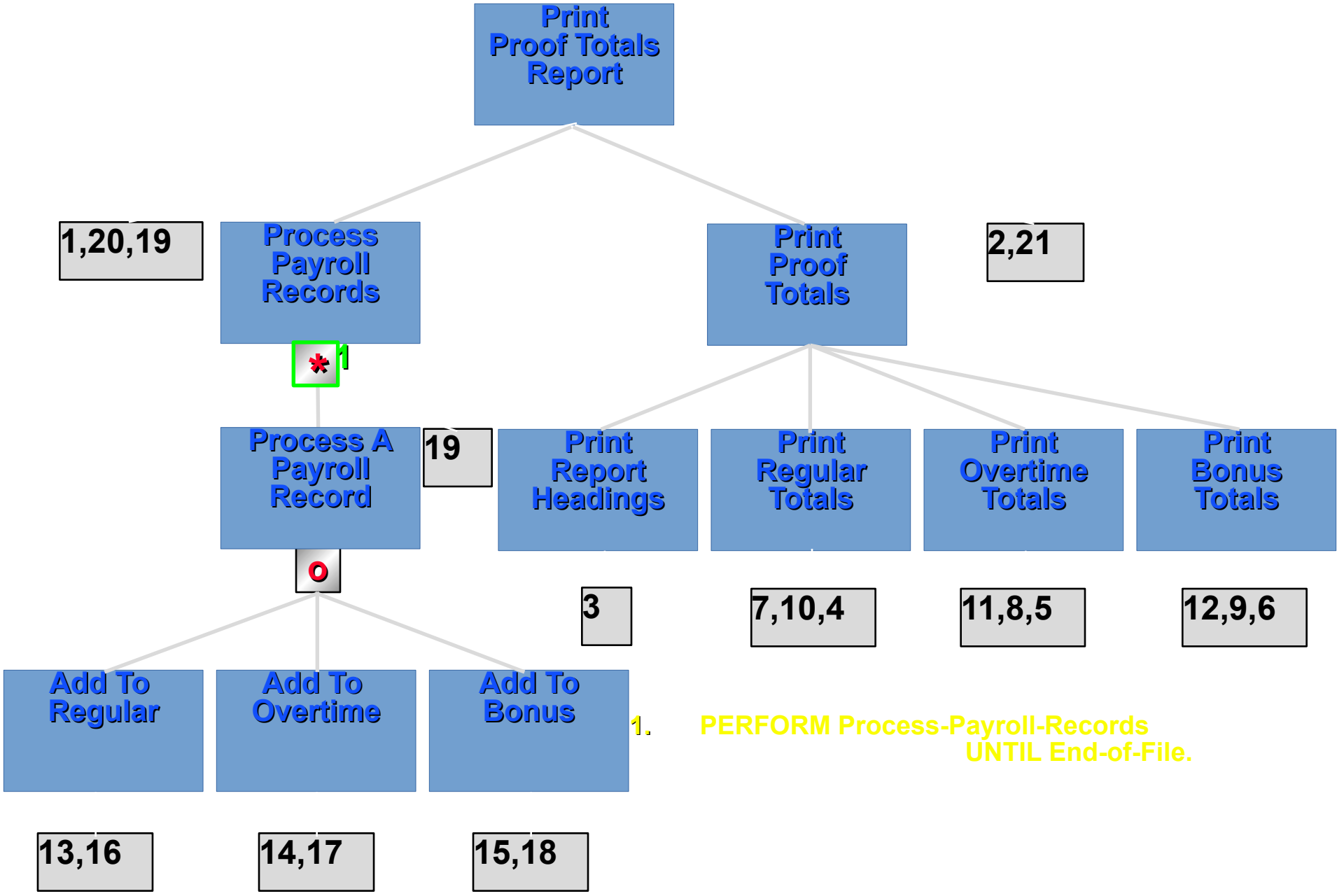
"What iteration statement must I use here to get the program to work correctly?".

Write the Iteration statement, give it a number and insert its number into the diagram.

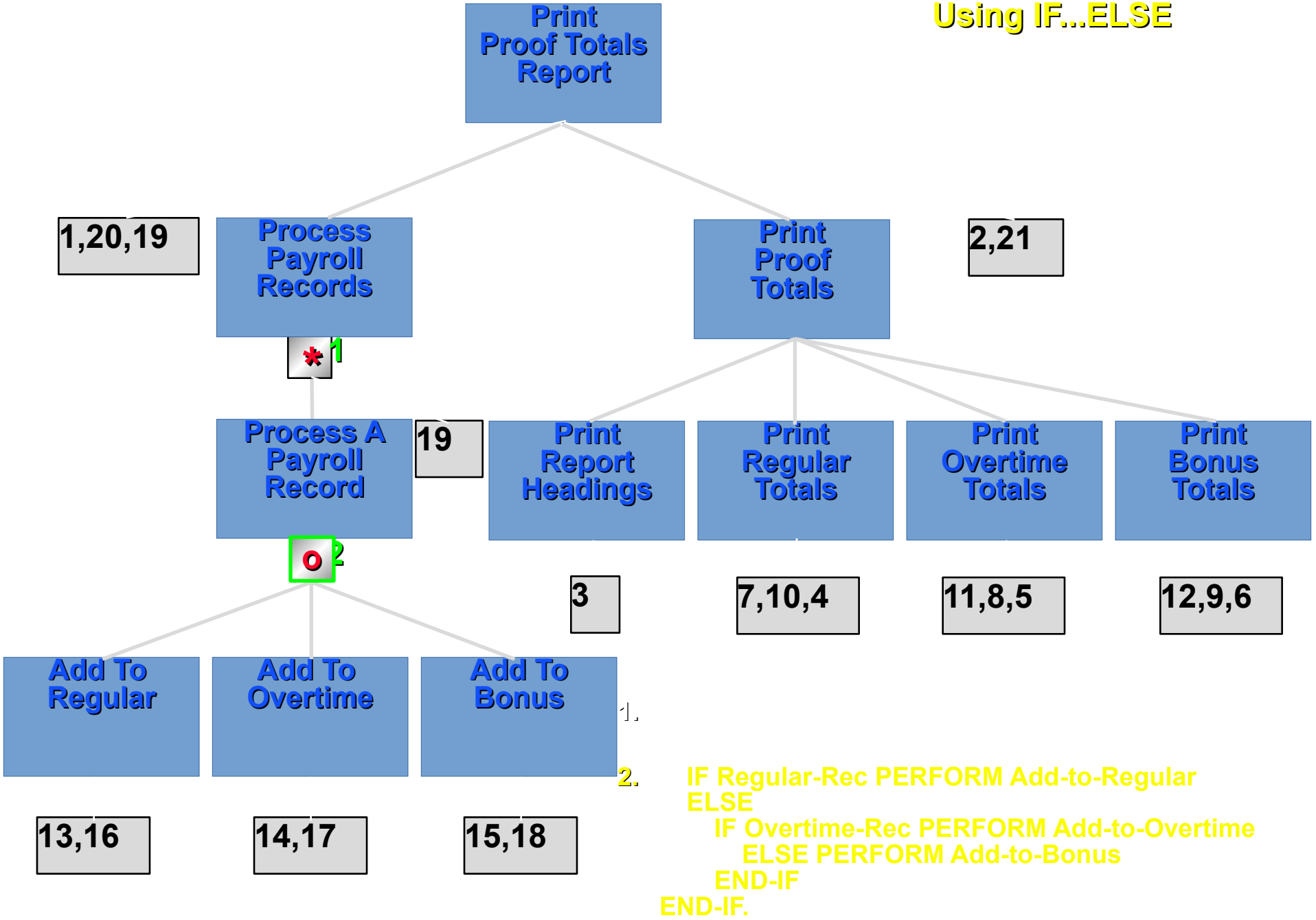
Go to each selection component and ask;

" What selection statement must I use here to get the program to work correctly?".

Write the Selection statement, give it a number and insert its number into the diagram.

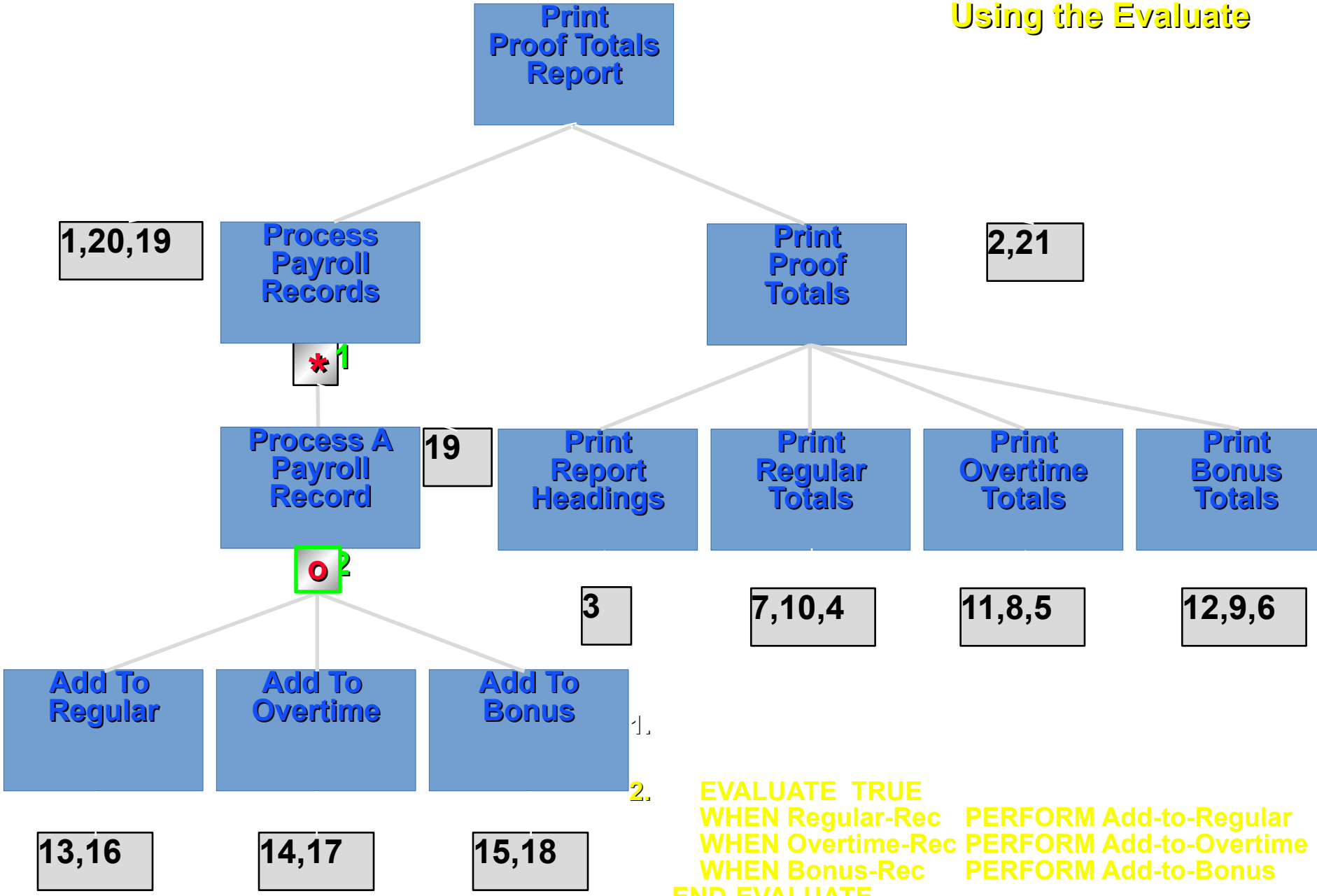

Print Proof Totals Report
1,20,19
Process Payroll Records
*
1
Print Proof Totals
2,21
Process A Payroll Record
19
Print Report Headings
Print Regular Totals
Print Overtime Totals
Print Bonus Totals
o
3
7,10,4
11,8,5
12,9,6
Add To Regular
Add To Overtime
Add To Bonus
1. PERFORM Process-Payroll-Records
UNTIL End-of-File.
13,16
14,17
15,18

Using IF...ELSE



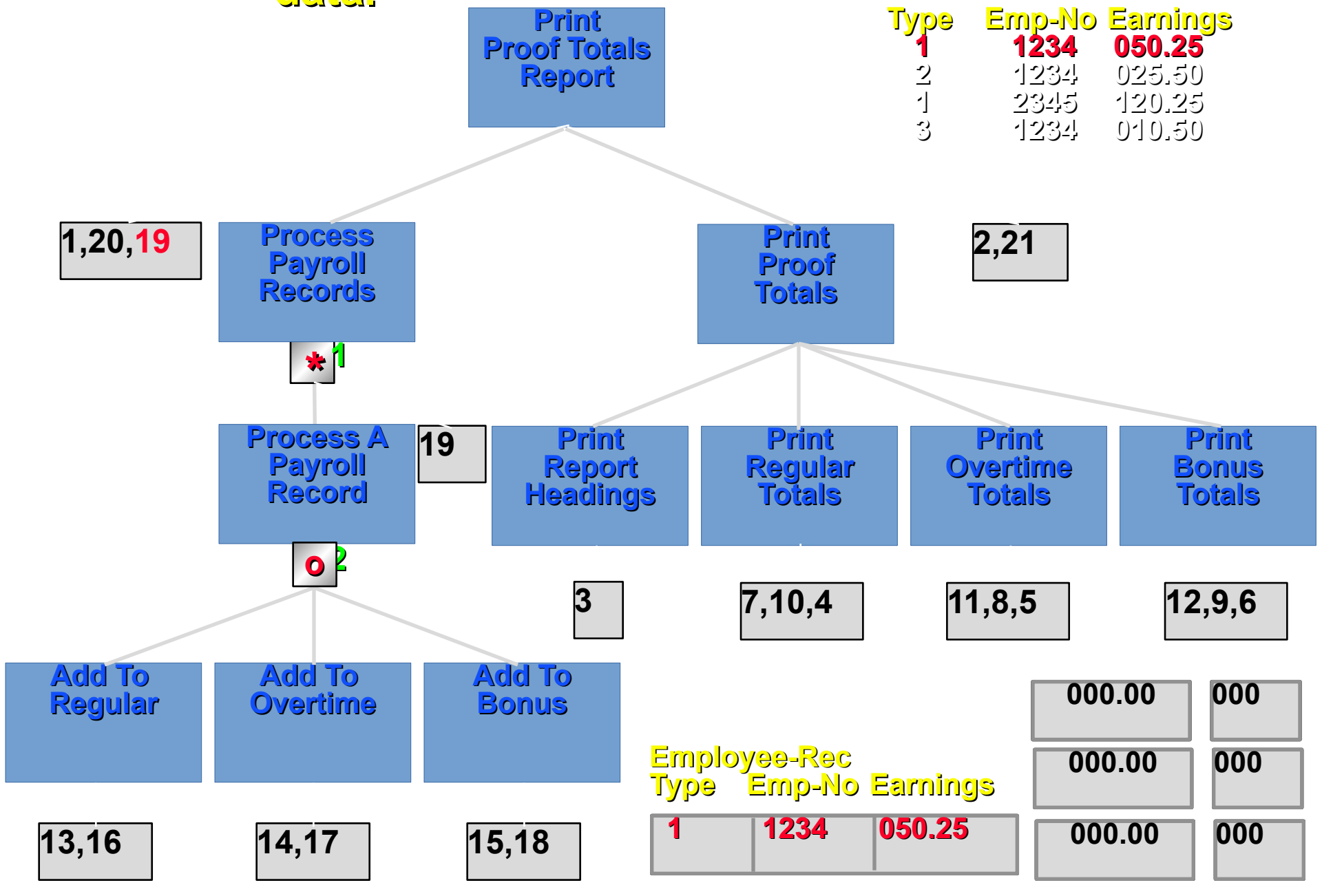

```
2.    IF Regular-Rec PERFORM Add-to-Regular
      ELSE
         IF Overtime-Rec PERFORM Add-to-Overtime
            ELSE PERFORM Add-to-Bonus
         END-IF
      END-IF.
```

Using the Evaluate




```
2.    EVALUATE  TRUE
      WHEN Regular-Rec    PERFORM Add-to-Regular
      WHEN Overtime-Rec PERFORM Add-to-Overtime
      WHEN Bonus-Rec      PERFORM Add-to-Bonus
END-EVALUATE.
```

7. Test the program using test data.



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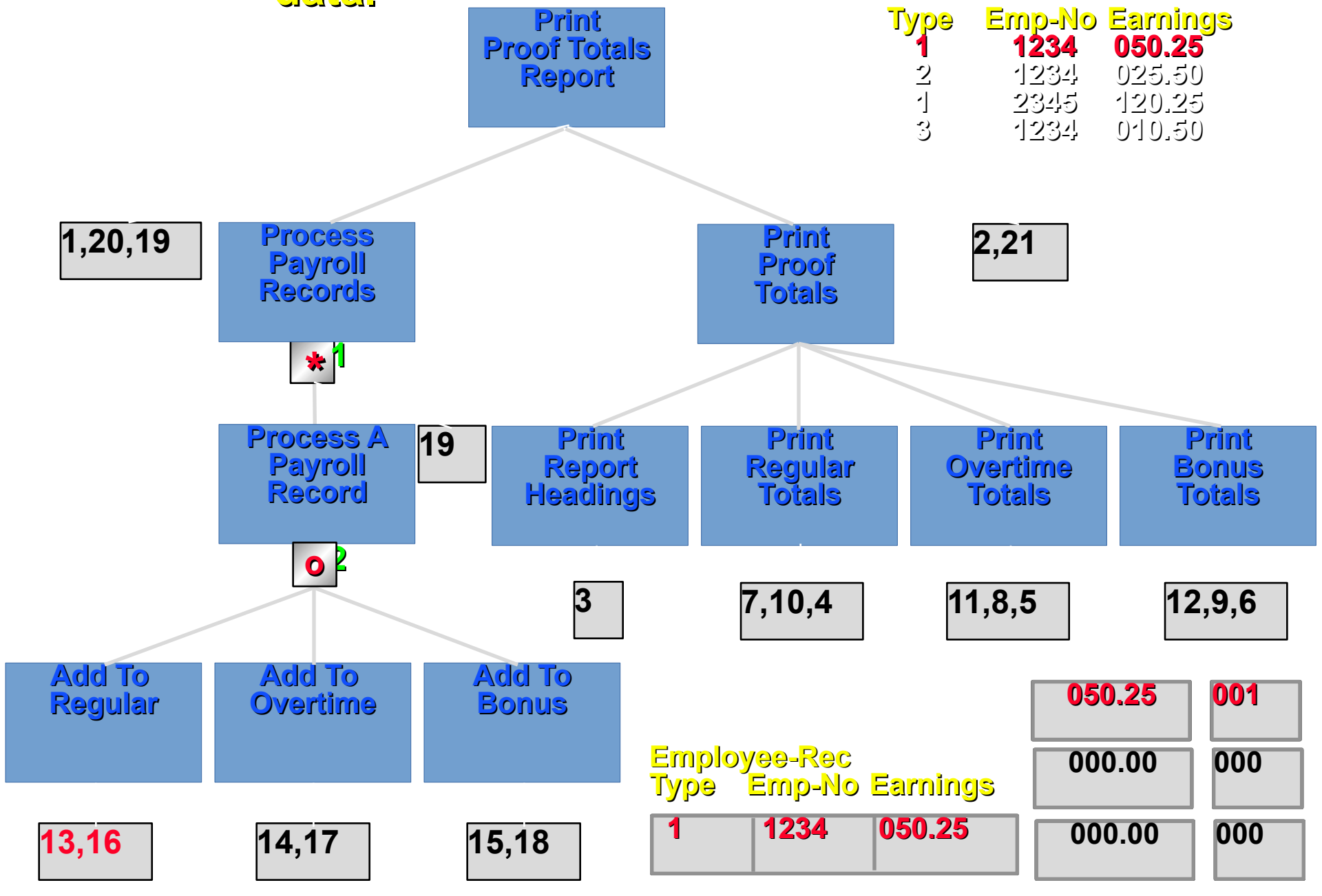
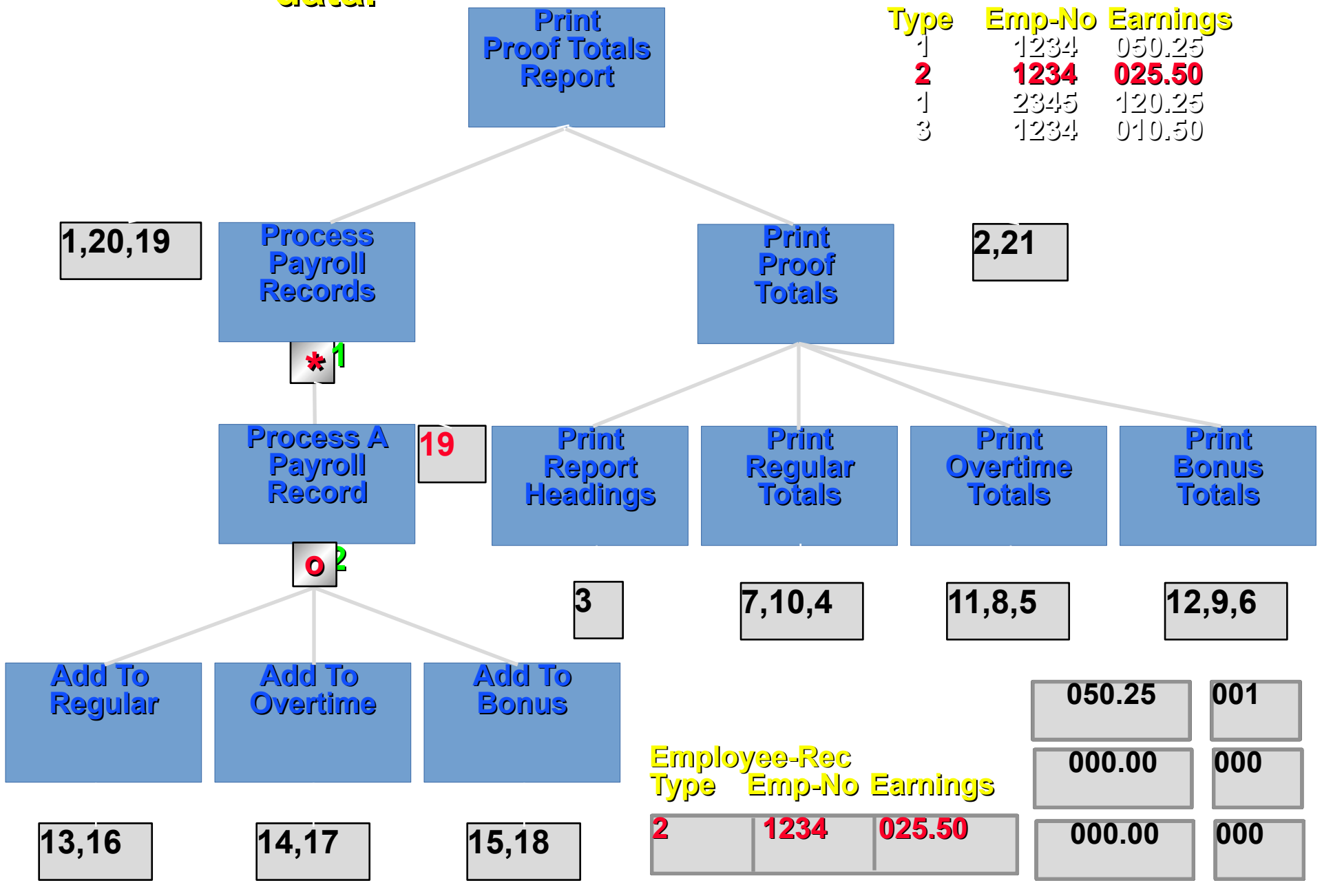
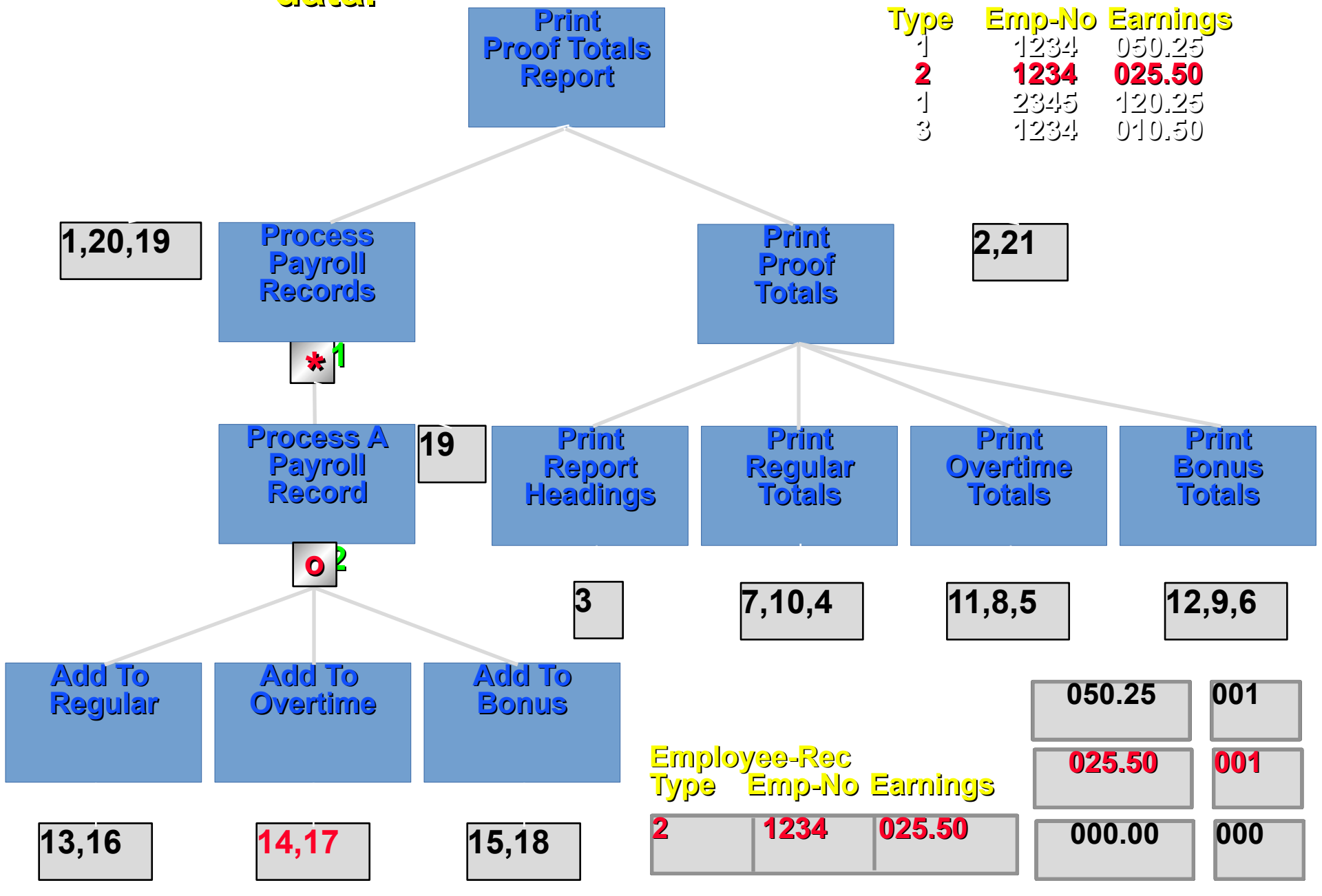


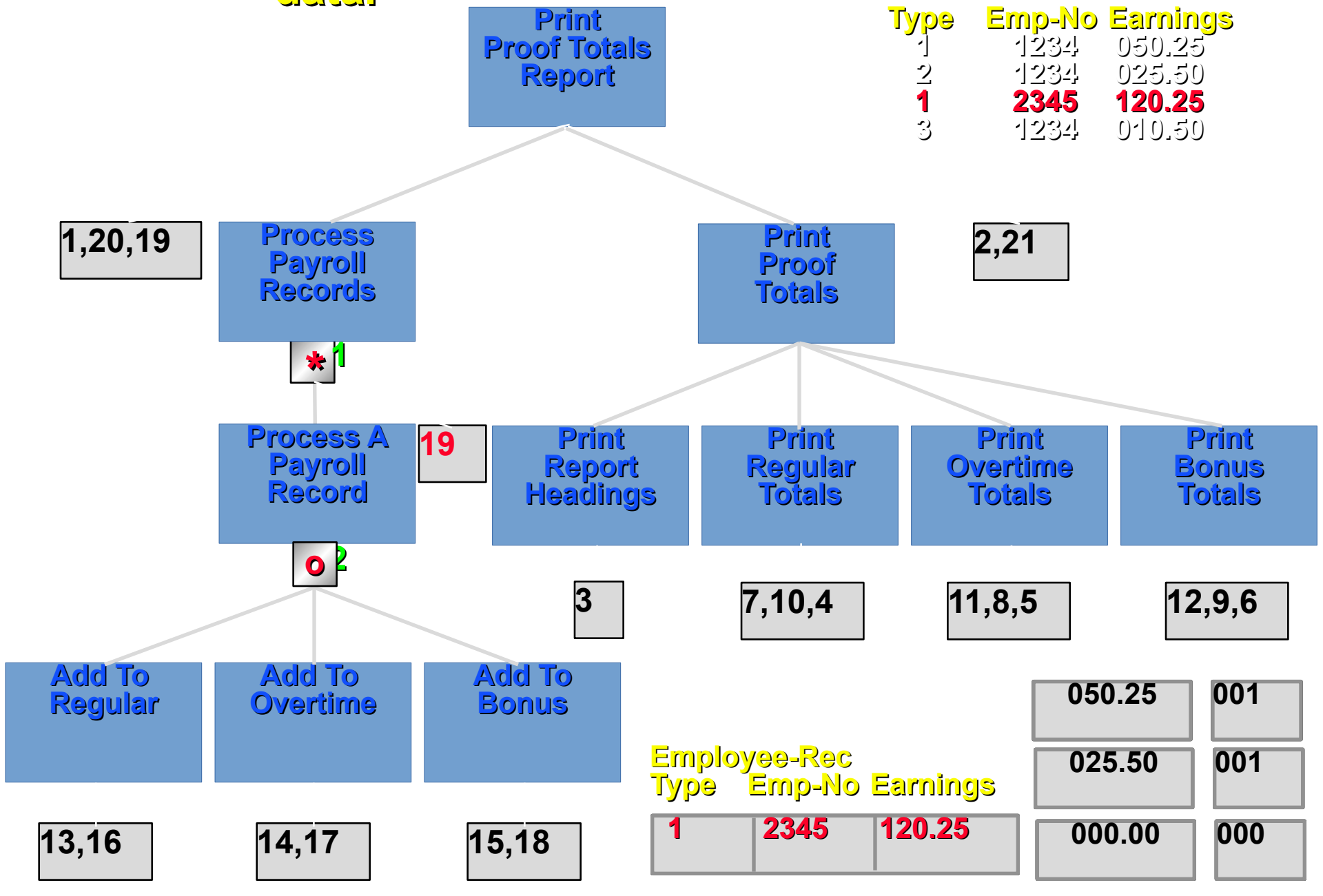
7. Test the program using test data.



7. Test the program using test data.



7. Test the program using test data.



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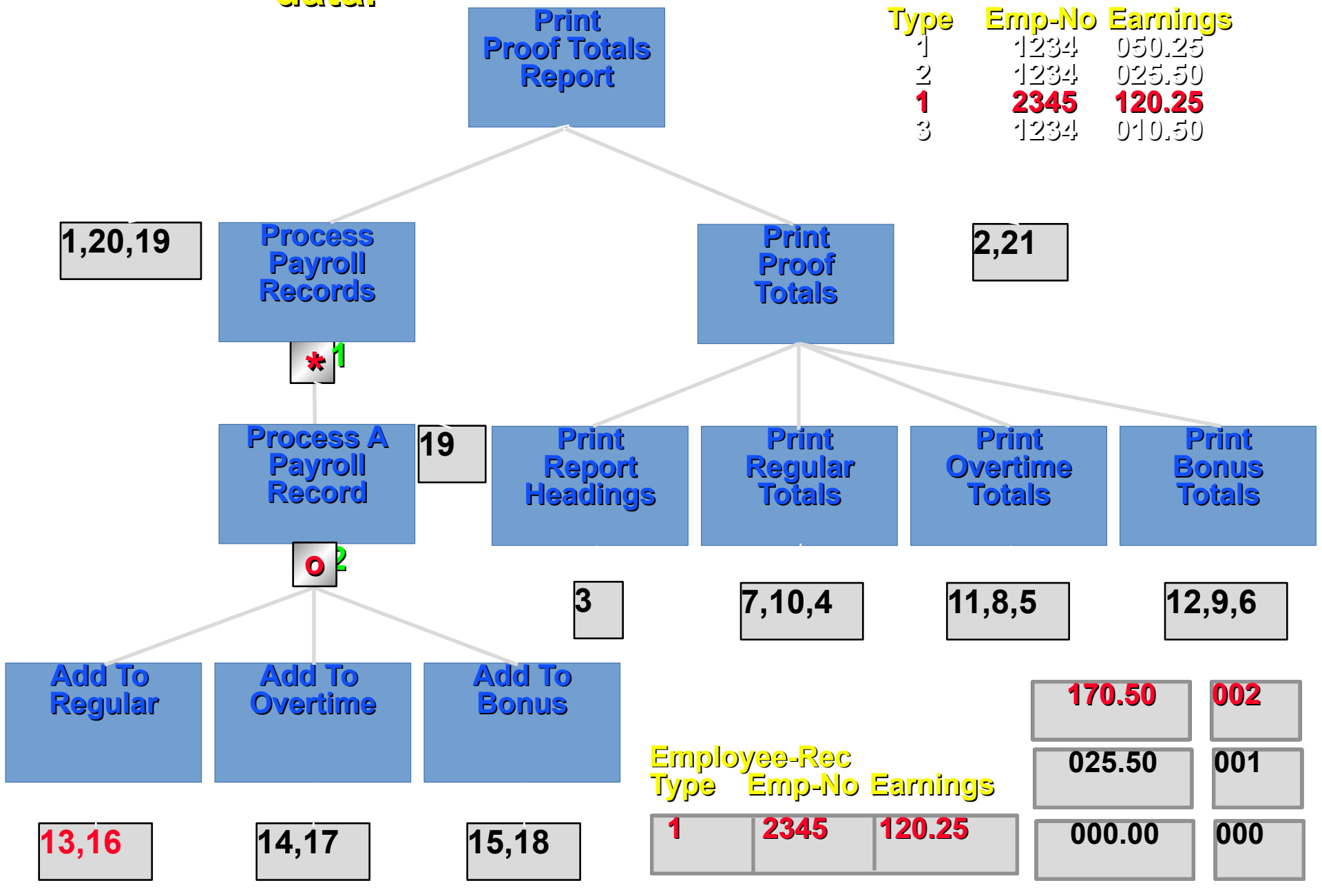
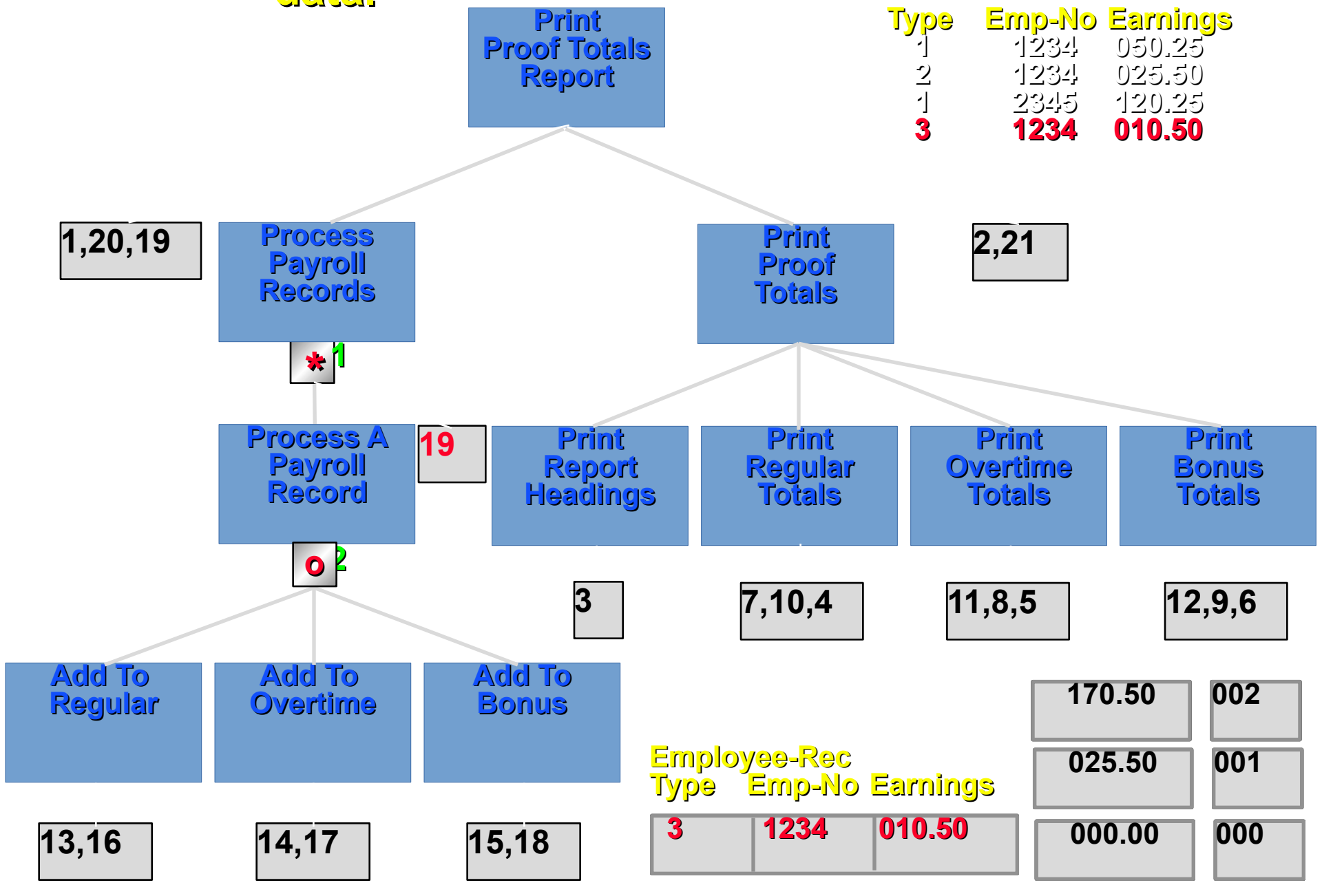
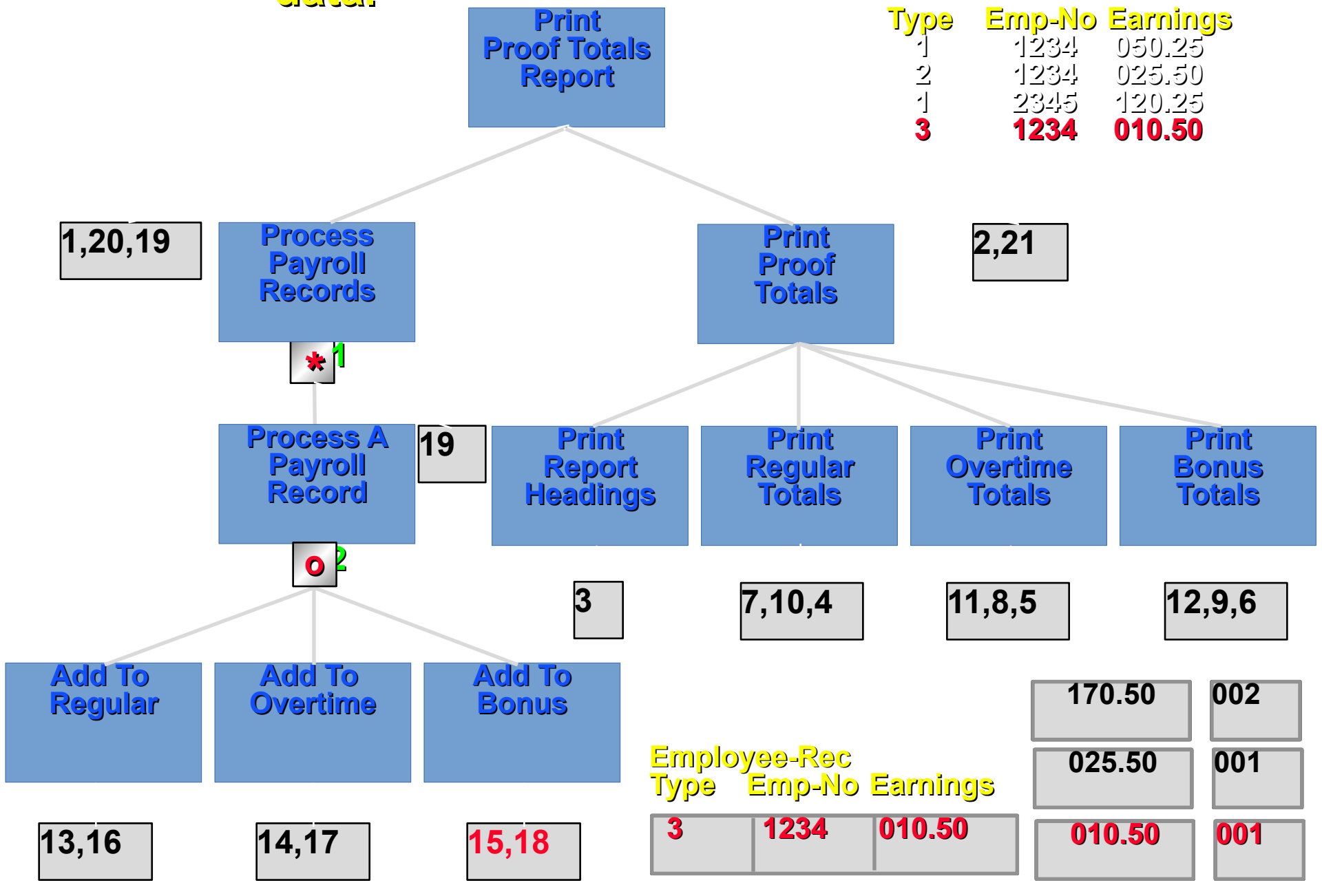


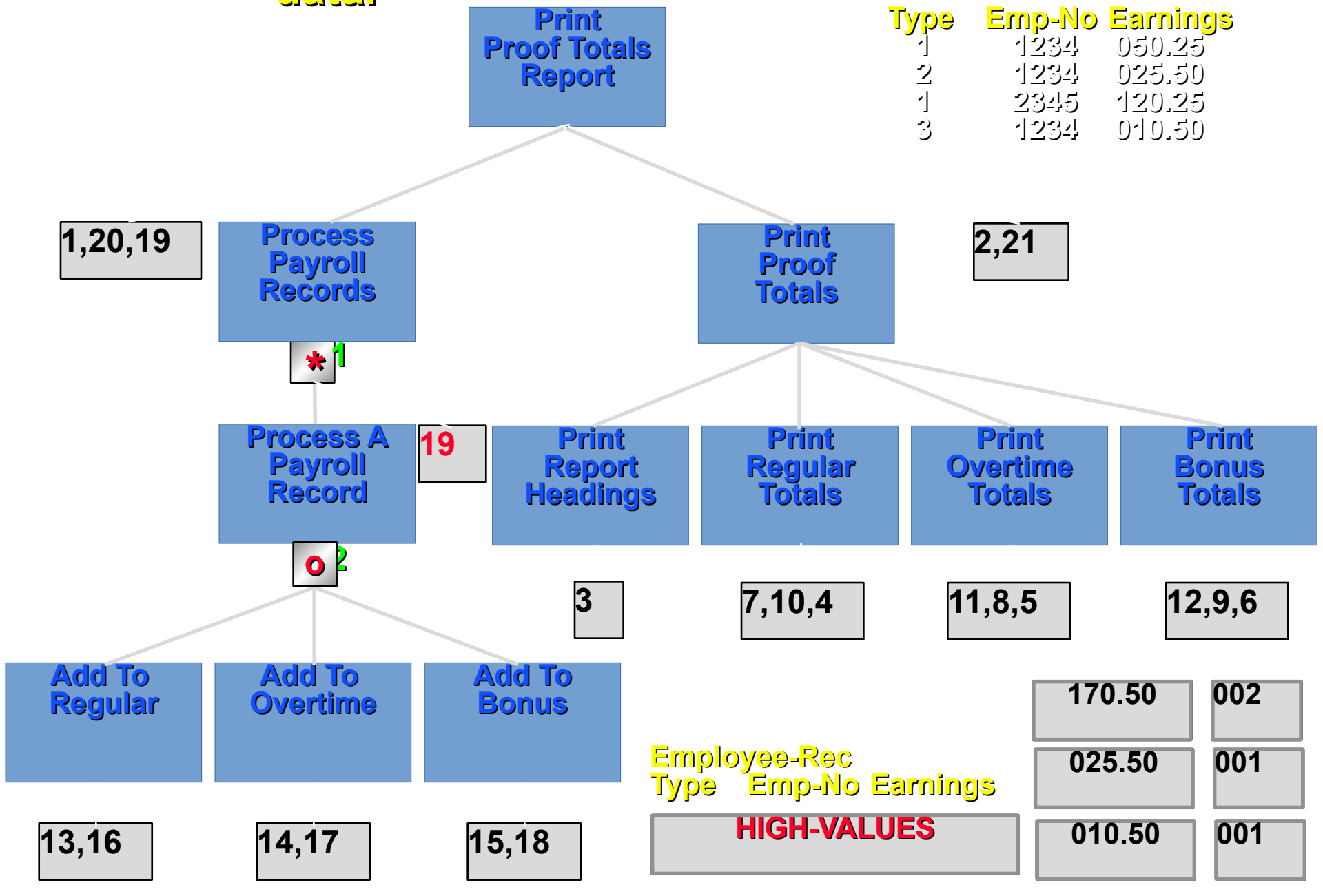
7. Test the program using test data.



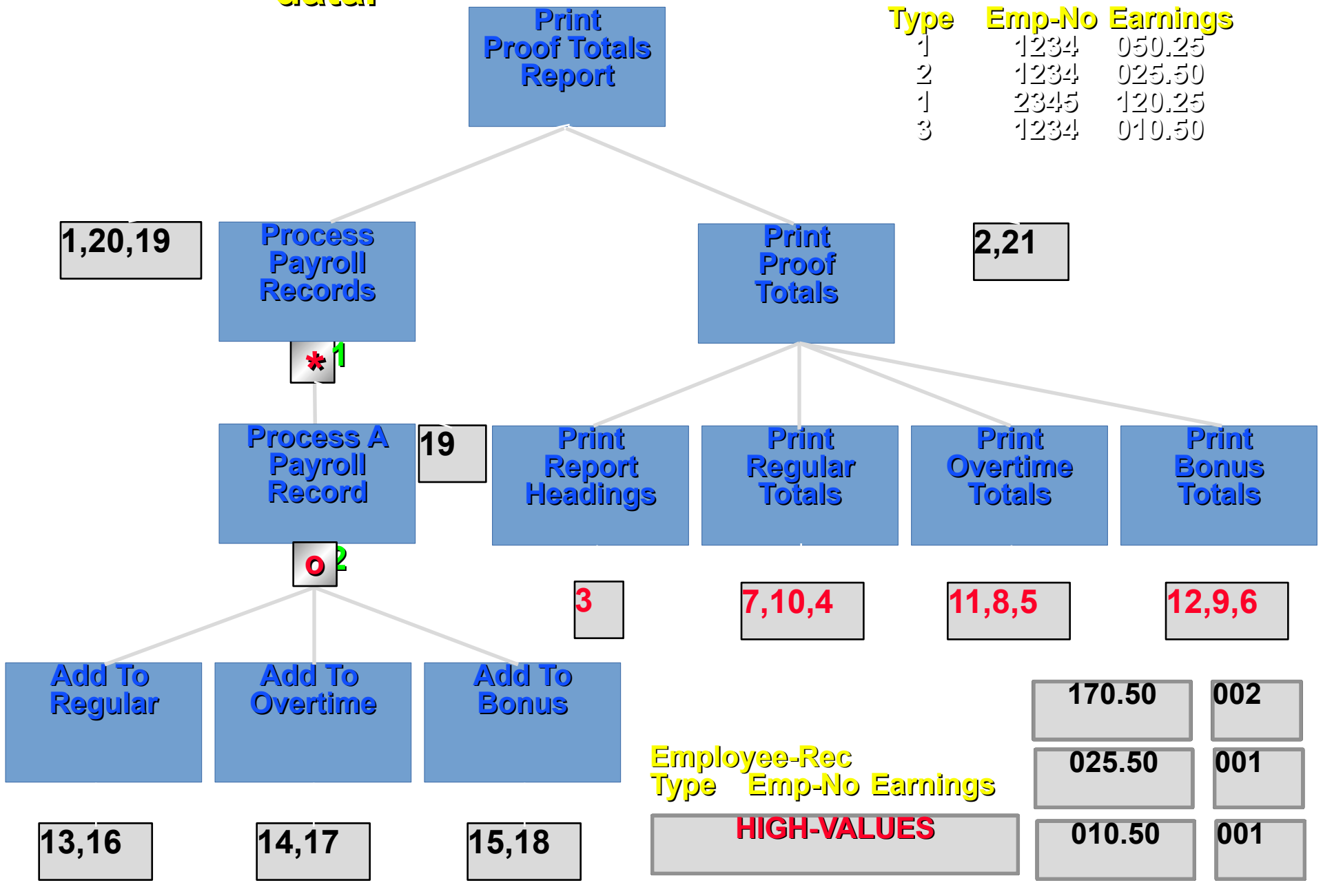
7. Test the program using test data.



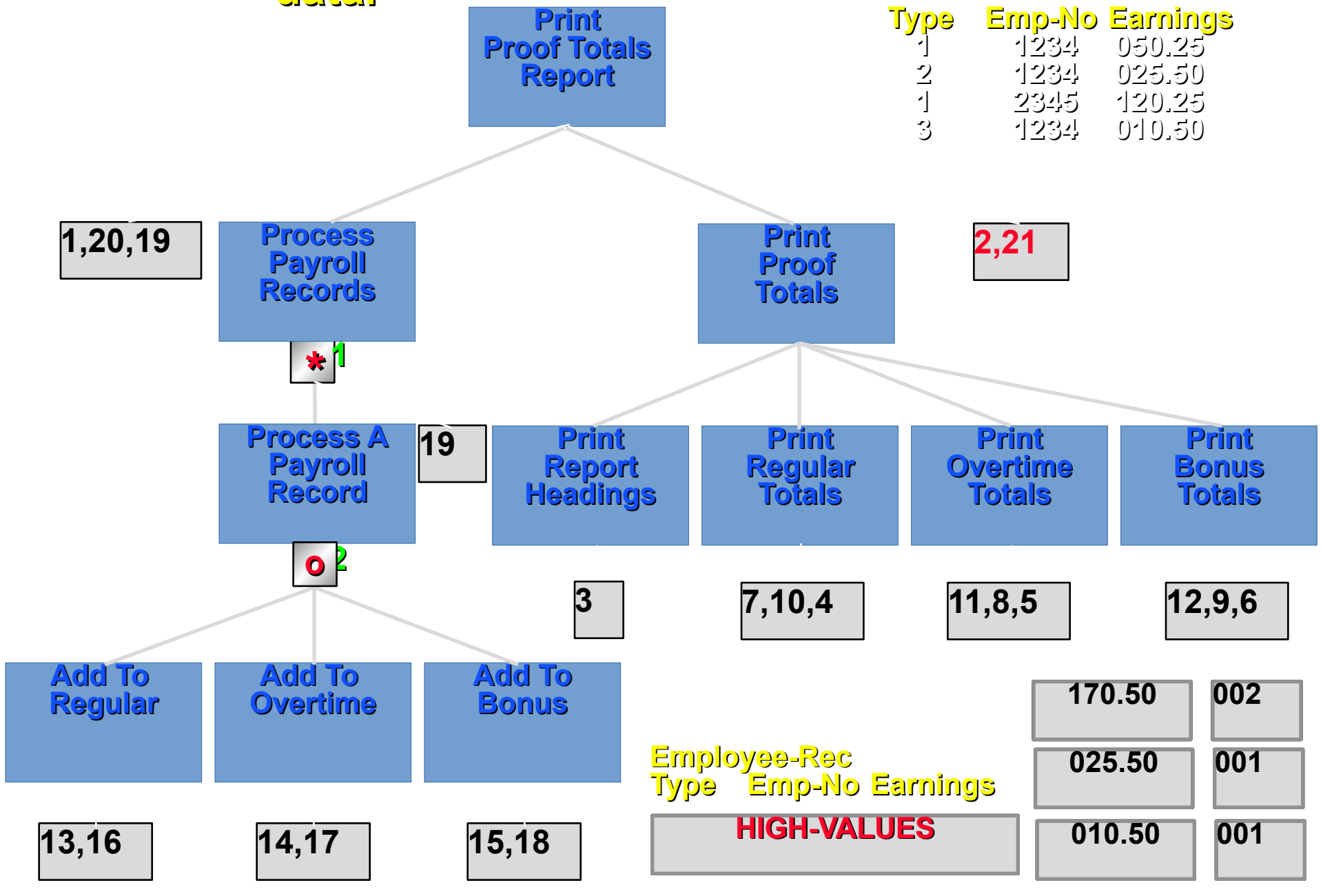
7. Test the program using test data.



7. Test the program using test data.



7. Test the program using test data.



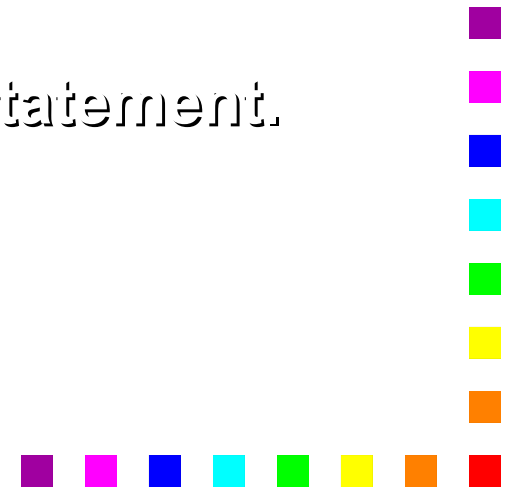
8. Translate the populated PSD into COBOL code

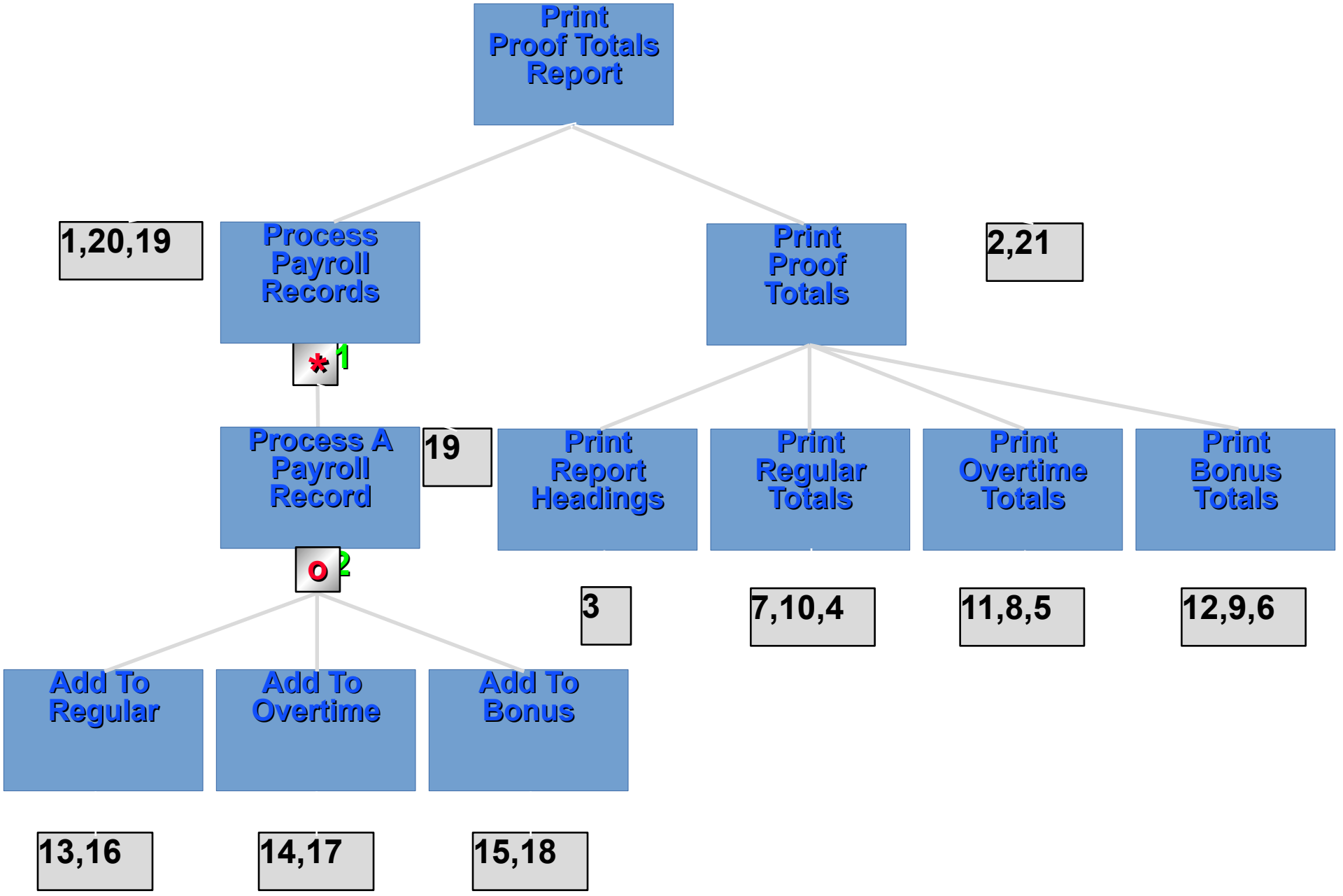
Start at the top of the diagram.

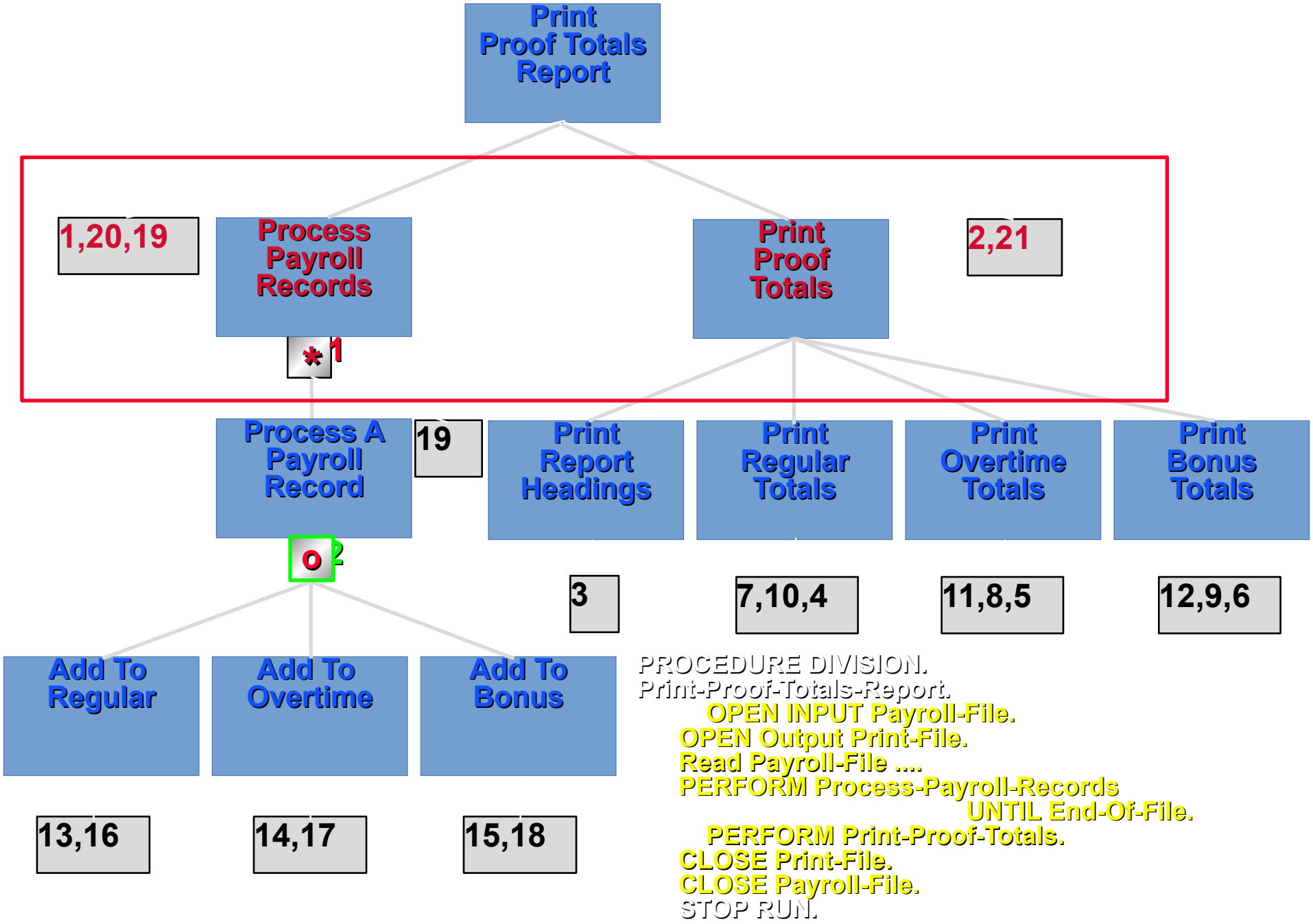
At each parent component write the sequence of executable operations, child components and iteration and selection conditions as statements in a paragraph.

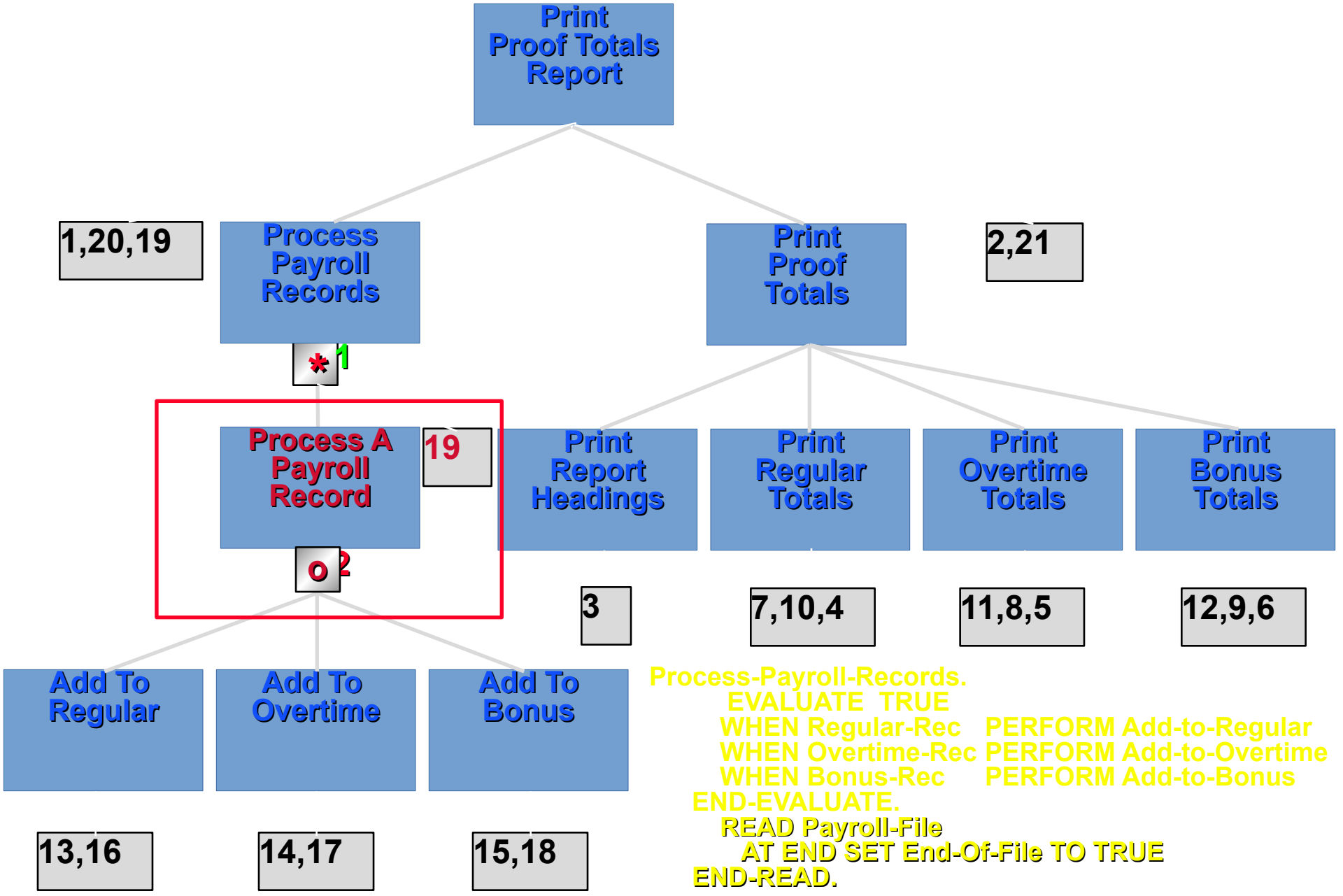
Replace child components with PERFORMs of the component name.

At the top component, use the name as the name of the first paragraph and preceed it with the PROCEDURE DIVISION header.
End the sequence with the STOP RUN statement.




Print Proof Totals Report
1,20,19
Process Payroll Records
*
1
Print Proof Totals
2,21
Process A Payroll Record
19
Print Report Headings
Print Regular Totals
Print Overtime Totals
Print Bonus Totals
o
2
3
7,10,4
11,8,5
12,9,6
Add To Regular
Add To Overtime
Add To Bonus
13,16
14,17
15,18


Print Proof Totals Report
1,20,19
Process Payroll Records
*
1
Print Proof Totals
2,21
Process A Payroll Record
19
Print Report Headings
Print Regular Totals
Print Overtime Totals
Print Bonus Totals
o
2
3
7,10,4
11,8,5
12,9,6
Add To Regular
Add To Overtime
Add To Bonus
13,16
14,17
15,18
PROCEDURE DIVISION.
Print-Proof-Totals-Report.
OPEN INPUT Payroll-File.
OPEN Output Print-File.
Read Payroll-File ....
PERFORM Process-Payroll-Records
UNTIL End-Of-File.
PERFORM Print-Proof-Totals.
CLOSE Print-File.
CLOSE Payroll-File.
STOP RUN.


Print Proof Totals Report
1,20,19
Process Payroll Records
*
1
Print Proof Totals
2,21
Process A Payroll Record
19
o
2
Print Report Headings
Print Regular Totals
Print Overtime Totals
Print Bonus Totals
3
7,10,4
11,8,5
12,9,6
Add To Regular
Add To Overtime
Add To Bonus
13,16
14,17
15,18
Process-Payroll-Records.
EVALUATE TRUE
WHEN Regular-Rec PERFORM Add-to-Regular
WHEN Overtime-Rec PERFORM Add-to-Overtime
WHEN Bonus-Rec PERFORM Add-to-Bonus
END-EVALUATE.
READ Payroll-File
AT END SET End-Of-File TO TRUE
END-READ.